



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
STA. MESA MANILA
COLLEGE OF ENGINEERING

Self-Study Report (SSR)

Bachelor of Science in Civil Engineering

Criterion 9

Continuous Quality Improvement

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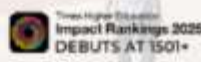
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CRITERION 9: CONTINUOUS QUALITY IMPROVEMENT

Introduction

The Polytechnic University of the Philippines (PUP) is committed to maintaining and enhancing the quality of its academic programs through a systematic Continuous Quality Improvement (CQI) process (Figure 9.1). This process ensures that the program remains responsive to the evolving needs of industry, the profession, and society while maintaining alignment with the University's vision, mission, and institutional goals. The CQI framework integrates regular assessment, evaluation, and feedback mechanisms to determine the extent to which the Program Educational Objectives (PEOs) and Student Outcomes (SOs) are achieved.

The University has established a structured process for assessing and evaluating the attainment of Student Outcomes. These outcomes describe the knowledge, skills, and professional attributes that students are expected to demonstrate upon graduation. Assessment is conducted using both (1) direct and (2) indirect methods, including course-embedded assessments, capstone design projects, laboratory performance, internship or practicum evaluations, and student portfolios. Indirect measures include graduating student surveys, alumni surveys, and employer feedback. The results of these assessments are systematically collected and analyzed by the departmental assessment committee and faculty members to determine the level of attainment of each student outcome.

The evaluation of Student Outcomes is carried out periodically through faculty deliberations, program review meetings, and curriculum committee discussions. These evaluations determine whether the performance indicators associated with each outcome meet the established performance targets. When gaps or areas for improvement are identified, the program formulates and implements appropriate corrective actions, which may include revising course content, improving teaching and learning strategies, updating laboratory facilities, and enhancing assessment tools. These actions are documented and monitored to ensure that improvements are effectively implemented and sustained.

In addition to student outcomes, the program implements a systematic process for assessing the attainment of Program Educational Objectives. The PEOs describe the expected accomplishments of graduates three to five years after graduation, including their professional practice, leadership roles, and contributions to society. The evaluation of PEO attainment is conducted through alumni tracer studies, employer surveys, industry consultations, and advisory board feedback. These sources provide valuable insights into graduates' professional performance and career progression, which serve as indicators of the program's effectiveness in achieving its educational objectives.

The results of the PEO evaluation are reviewed by the program faculty and relevant stakeholders to determine whether the program objectives remain relevant and aligned with current industry and societal needs. Based on these evaluations, the program may implement improvements such as curriculum revisions, integration of emerging technologies, strengthening of research and extension activities, and enhancement of industry partnerships. These actions ensure that the program continues to produce graduates who are competent, innovative, and socially responsible professionals.

The Continuous Quality Improvement process in PUP is supported by institutional structures such as the Quality Assurance Center (QAC) and program-level committees responsible for outcomes-based education implementation. These units facilitate data collection, analysis, and documentation of assessment results and improvement initiatives. The CQI cycle follows a Plan–Do–Check–Act (PDCA) approach, ensuring that assessment findings are translated into actionable improvements and that the impact of these improvements is subsequently evaluated.

Through this systematic and evidence-based approach, the Polytechnic University of the Philippines ensures that the attainment of Student Outcomes and Program Educational Objectives is continuously monitored and enhanced. The integration of assessment results into program planning and decision-making demonstrates the University's strong commitment to continuous quality improvement, academic excellence, and the development of globally competitive graduates.

A systematic approach that allows the program to address the need for a specific program improvement and evidence of student learning has become a requirement demanded by stakeholders for program accountability through the idea of Continuous Quality Improvement (CQI) and its implementation. When CQI is used in engineering education, program and classroom effectiveness are the main concerns, with an emphasis on student learning. Through assessment activities, the program can look at the data to determine what is working and what needs to be improved. Figure 9.1 shows the PUP Continuous Quality Improvement (CQI) Model.

The Continuous Quality Improvement (CQI) model outlined for the Polytechnic University of the Philippines (PUP) Civil Engineering Department, using the Plan-Do-Check-Act (PDCA) approach, plays a crucial role in meeting the Philippine Technological Council's (PTC) Accreditation and Certification Board for Engineering and Technology (ACBET) standards under Criterion 9. This model ensures ongoing quality monitoring and improvement processes within the department, aligning with the accreditation requirements. Here's a detailed discussion on each stage of the PDCA cycle:

1. Plan Stage

The planning phase focuses on defining quality objectives for the Civil Engineering program, establishing indicators aligned with the Program Educational Objectives (PEOs) and Student Outcomes (SOs), and setting up necessary resources.

During this stage, the department collaborates with stakeholders, including faculty, industry partners, and alumni, to identify areas requiring improvement. PTC accreditation mandates that educational objectives align with industry expectations and global standards, so the planning phase emphasizes defining metrics that reflect both PEOs and SOs.

Formulation of measurable PEOs and SOs; defining assessment metrics; developing a timeline and resource allocation plan; setting baseline data for comparison during evaluations. These metrics serve as benchmarks for later evaluation stages to ensure continuous alignment with PTC's educational standards.

2. Do Stage

The "Do" phase involves implementing the planned strategies and processes to achieve the program's educational objectives and student outcomes.

Activities in this stage include curriculum implementation, instruction delivery, laboratory and fieldwork execution, and provision of student services. Faculty members engage in curriculum delivery, while students participate in experiential learning and projects designed to meet the SOs.

The department executes educational and administrative processes, collects data from assessments, and document activities. Here, the Civil Engineering Department ensures all processes are conducted per established standards and that all instructional methods and resources are effectively utilized. The documentation provides foundational data for the "Check" phase.

3. Check Stage

This stage evaluates the extent to which the PEOs and SOs are being achieved by analyzing the data collected in the "Do" phase.

The department assesses student performance through evaluations, including exams, project reports, and capstone presentations. Faculty assessments, student feedback, and alumni input are also analyzed to gauge how well the program meets its objectives.

The department measures performance against defined benchmarks, analyze feedback, and document findings in detailed reports. The Civil Engineering Department needs to demonstrate that the data reflects a thorough evaluation of program goals. The Check stage provides evidence-based insights into areas needing attention, forming the foundation for data-driven decisions.

4. Act Stage

This stage addresses issues identified in the Check phase by implementing corrective actions to improve program quality.

Based on the evaluation results, the department adjusts curricula, teaching methods, or resource allocations. Actions might include revising course content, updating lab equipment, or providing faculty development workshops. These improvements aim to address gaps in achieving PEOs and SOs.

The department implements improvements, documents changes, and prepare for reassessment in the next CQI cycle. Each adjustment is recorded as part of the CQI documentation to demonstrate a commitment to continuous quality improvement, to document changes and their impact.

The PDCA-based CQI model applied in the PUP Civil Engineering Department is tailored to align with PTC-ACBET's accreditation requirements. This approach provides a systematic and evidence-based process for ongoing quality assessment and improvement. Each phase of the PDCA model ensures that educational objectives are continuously evaluated and improved upon, meeting the high standards set forth for engineering education.

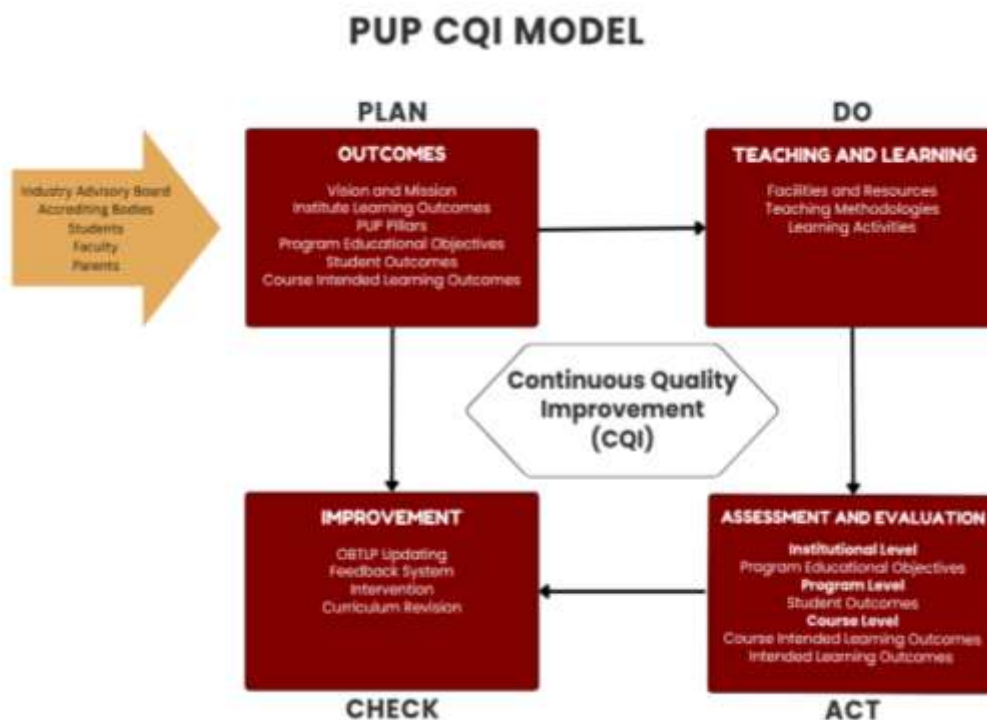


Figure 9.1 PUP CQI Model based on P-D-C-A (Plan – Doc – Check – Act)

A. Program Educational Objectives

1. A listing and description of the assessment processes used to gather the data upon which the evaluation of each program educational objective is based. Examples of assessment processes may include, but are not limited to, employer surveys, graduate surveys, focus groups, industrial advisory committee meetings, or other processes that are relevant and appropriate to the program.
2. A listing and description of the evaluation processes used for each of the program educational objectives.
3. The frequency with which these assessment and evaluation processes are carried out.
4. The expected satisfactory level of attainment for each of the program educational objectives. This should include report to show level of attainment.
5. Actual level of attainment versus the expected the satisfactory level for each program educational objectives.
6. Continuous improvement actions undertaken and any significant future improvement plans.
7. How the results are documented and maintained. Provide samples of results.

The Polytechnic University of the Philippines (PUP) implements a systematic process for assessing and evaluating the attainment of its Program Educational Objectives (PEOs) as

part of its commitment to Continuous Quality Improvement (CQI) and compliance with the outcomes-based quality assurance framework of the Philippine Technological Council (PTC). The Program Educational Objectives describe the expected professional achievements of graduates three to five years after graduation, reflecting their career advancement, professional competence, leadership, ethical responsibility, and contributions to society.

To determine the extent to which these objectives are achieved, the program employs a variety of assessment processes that gather relevant data from key stakeholders, including alumni, employers, industry partners, and faculty members. These processes provide both quantitative and qualitative evidence regarding the professional performance and development of graduates in their respective fields.

Among the primary assessment tools used by the program is the Alumni or Graduate Survey, which collects information from graduates several years after completing the program. The survey gathers feedback on career progression, professional achievements, continuing education, leadership roles, and the relevance of the competencies acquired during their studies. The results of the survey provide valuable insights into how effectively the program prepared graduates for professional practice.

Another important source of assessment data is the Employer Survey, which solicits feedback from organizations that employ program graduates. Employers are asked to evaluate graduates in areas such as technical competence, problem-solving ability, teamwork, communication skills, professional ethics, and leadership potential. The information gathered from employers helps determine whether the program is producing graduates who meet the expectations and standards of industry and the profession.

The program also benefits from Government-Industry-Academe Board (GIAB) Committee meetings, where representatives from industry, professional organizations, academic leaders, faculty, and other stakeholders provide recommendations on curriculum relevance, emerging technologies, and professional competencies required in the workplace. These consultations ensure that the Program Educational Objectives remain aligned with current and future industry needs.

In addition, the program conducts focus group discussions with alumni, employers, and other stakeholders to gather deeper insights regarding the professional development of graduates and the effectiveness of the curriculum in preparing them for their careers. These discussions often highlight strengths of the program as well as areas where improvements may be necessary.

The data collected from these various assessment processes are systematically analyzed by the program faculty and the departmental curriculum and assessment committees. The results are then discussed during faculty meetings and program review sessions to evaluate the degree to which the Program Educational Objectives are being achieved. When areas for improvement are identified, the program implements appropriate corrective actions, such as curriculum enhancement, strengthening of industry collaboration, or integration of new technologies and professional practices.

Through these structured assessment mechanisms, the Polytechnic University of the Philippines ensures that the Program Educational Objectives remain relevant, measurable, and aligned with stakeholder expectations. The continuous analysis and utilization of assessment results demonstrate the University's strong commitment to maintaining academic excellence and fostering the professional success of its graduates through an effective Continuous Quality Improvement process.

As stated in Criterion 1, the PEOs for the Civil Engineering program were developed with the university's mission and vision, considering feedback from members of the Government Industry Academe Board (GIAB), including employers, alumni, and members of an accredited professional organization. Additional feedback was obtained from internal stakeholders, including teachers and students. Three to five years after graduation, PEOs were developed to define expected graduate attributes. To ascertain whether the graduate cohort met the Performance Indicators for each PEO, the PEO assessment will be carried out by employer survey or interview. Alumni are expected to complete at least one PI for each of the three PEOs, which have a minimum of three PIs each. If 60% or more of the graduating class achieves every PEO, then the PEOs are deemed accomplished. If the outcome for several cohorts is much below the benchmarked PI, a program-level continuous quality improvement (CQI) activity must be carried out. A sample of PEO survey form is presented in Figure 9.2 ([Attachment C9-01 – PEO Survey Form](#)).



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
Vice President for Academic Affairs
College of Engineering

ASSESSMENT AND EVALUATION OF PROGRAM EDUCATIONAL OBJECTIVES

☐ Employers ☐ Alumni

The purpose of this form is to collect opinions and observations to help us evaluate whether our Program Educational Objectives have been met by our graduates.

Name of Graduate: _____ **Program:** _____
Class Graduated: _____

The Program Educational Objectives describe the career and professional accomplishment the program has prepared the graduates to achieve within three to five years after graduation. A Bachelor of Science in Civil Engineering alumni shall be:

- PEO 1. Demonstrating Engineering Profession:** Employed and/or practiced in the field of Civil Engineering, such as but not limited to Construction Engineering and Management, Geotechnical Engineering, Structural Engineering, Transportation Engineering, and Water Resources Engineering, or other related fields.
- PEO 2. Social and Ethical Responsibility:** Committed members of professional and other allied organizations engaging in community development or nation-building.
- PEO 3. Lifelong Learning:** Involved in continuous training and development in current trends and advancements in their fields of specialization.

Please evaluate the attainment of the Program Educational Objectives of our graduate by putting a "✓" on the appropriate scale:

1 – No Extent at All 2 – Low Extent 3 – Medium Extent 4 – Great Extent 5 – Very Great Extent

Figure 9.2 PEO Survey Form

As part of the department's commitment to continuous quality improvement, a systematic process is implemented to assess the attainment of the Program Educational Objectives (PEOs). The evaluation of the PEOs is anchored on data gathered from the graduates and employers, whose feedback provides evidence on how well the program prepares its students for professional practice and career development after graduation.

The assessment process for each PEO is based primarily on graduate surveys and employer surveys. These instruments are designed to gather relevant information on the graduates' professional growth, competencies, workplace performance, and alignment of their achievements with the intended program educational objectives. The surveys serve as an important mechanism for determining whether the graduates, within a few years after completing the program, have demonstrated the attributes and outcomes embodied in the PEOs.

For the current assessment cycle, the respondents consisted of graduates under the 2018 curriculum who completed the program in 2022. In accordance with the department's assessment timeline, the PEO survey was administered in 2025, or three years after graduation, which is considered an appropriate period for graduates to have gained sufficient

academic, professional, and workplace experience to meaningfully evaluate the extent to which the PEOs have been attained.

To facilitate efficient and wider data collection, the department deployed the survey instruments through Google Forms, with the survey links disseminated via email and other online platforms. This approach enabled the department to reach graduates and employers conveniently and increased accessibility for respondents across different locations. The use of online survey tools also allowed for organized data gathering, easier monitoring of responses, and efficient consolidation of results ([Attachment C9-02 – Screenshot of the Google Link Form](#)).

After the survey administration, the collected data were tabulated, evaluated, and analyzed by the department. The analysis focused on determining the level of attainment of each PEO based on the responses received from the identified stakeholders. The findings provided evidence-based inputs for reviewing the effectiveness of the curriculum, student preparation, and overall program implementation. Any identified strengths and areas needing improvement were considered in the department's continuous quality improvement initiatives.

The Program Educational Objectives (PEOs) serve as a crucial representation of the institution's vision for its graduates' careers three to five years after they finish their degrees. They also reflect the aims and goals of the college as well as the mission, vision, and basic values of the institution. As a result, there needs to be a solid procedure in place for program assessment and evaluation of educational objectives. The Polytechnic University of the Philippines - College of Engineering's BS Civil Engineering program uses evaluation procedures that are explained in Figure 9.3. The CQI committee receives the PEO assessment results and an evaluation report, same as in the SOs process.

The Program Educational Objectives (PEOs) articulate the envisioned accomplishments of BS Civil Engineering (BSCE) alumni within three to five years post-graduation, aligning with the institution's vision, mission, and core values. These objectives reflect the professional competencies, ethical standards, and societal contributions expected of PUP's graduates and directly support the college's overarching mission to develop skilled, responsible, and globally competent engineers. Given the significance of these objectives, the PUP College of Engineering implements a structured and robust evaluation process to ensure that each PEO remains relevant and aligned with the evolving demands of the engineering profession, the needs of industry, and the expectations of various stakeholders.

Evaluation Procedures for Program Educational Objectives

A structured and systematic process is implemented to evaluate the attainment of the Program Educational Objectives (PEOs) as part of the Continuous Quality Improvement (CQI)

system. These evaluation processes are aligned with the outcomes-based education framework required by the Philippine Technological Council and ensure that the program remains responsive to the needs of stakeholders, industry developments, and societal expectations.

The evaluation of Program Educational Objectives focuses on analyzing the data gathered from various assessment processes to determine whether graduates are achieving the intended professional accomplishments three to five years after graduation. The program utilizes several evaluation mechanisms to interpret the assessment results and determine the effectiveness and relevance of the PEOs.

One of the primary evaluation processes is the analysis of alumni tracer study results. The tracer study provides information on the employment status, career progression, professional achievements, and leadership roles of graduates. The program faculty review and analyze the data to determine whether graduates demonstrate the competencies, professional growth, and societal contributions described in the Program Educational Objectives.

Another important evaluation process involves the review of employer survey results. Feedback from employers regarding the performance of graduates in the workplace is analyzed to determine whether the program effectively prepares students with the required knowledge, technical skills, professional attitudes, and ethical responsibilities expected in the profession.

The program also conducts Government-Industry-Academe Board (GIAB) consultations and meetings, where industry representatives review the current Program Educational Objectives and provide recommendations based on emerging technologies, workforce requirements, and professional standards. The insights from these consultations are carefully evaluated to determine whether revisions to the PEOs or program components are necessary.

In addition, the program holds faculty program review meetings and curriculum committee deliberations where the collected assessment data are presented and discussed. During these meetings, faculty members evaluate whether the Program Educational Objectives remain relevant and aligned with the institution's vision and mission as well as the needs of stakeholders. The discussions also focus on identifying areas where program improvements can be implemented.

The results of these evaluation processes are documented and used to support decision-making related to program improvement. When the evaluation indicates that certain Program Educational Objectives are not being adequately achieved or require updating, the program undertakes appropriate actions such as revising the curriculum, enhancing industry

partnerships, strengthening research and extension activities, or refining the Program Educational Objectives themselves.

Through these systematic evaluation processes, the Polytechnic University of the Philippines ensures that the Program Educational Objectives are regularly reviewed, validated, and improved, thereby sustaining the relevance and quality of the program and supporting the professional success of its graduates.

The evaluation of PEOs involves a systematic approach outlined in Figure 9.3. This approach includes data collection, analysis, and review phases, which inform the continuous improvement cycle for the program's educational objectives. The Continuous Quality Improvement (CQI) Committee is tasked with overseeing these procedures, conducting regular assessments, and preparing detailed evaluation reports to guide program improvements. The assessment and evaluation processes comprise several key steps: ([Attachment C9-03 – The CQI Committee](#))

1. Stakeholder Feedback Collection

The evaluation process begins by collecting comprehensive feedback from stakeholders, which includes alumni, employers, industry experts, faculty, and advisory board members. Alumni surveys, employer feedback, and periodic industry consultations serve as primary tools in understanding how well graduates are meeting the expectations set forth in the PEOs. This data highlights the program's strengths and areas needing enhancement.

2. Performance Metrics and Indicators

To quantitatively assess each PEO, specific performance indicators are developed. These indicators are designed to measure the graduates' achievements in targeted areas such as Construction Engineering and Management, Structural Engineering, Transportation Engineering, Water Resources Engineering, and Geotechnical Engineering, among others. For example, PEO 1, which addresses career progression within specialized fields, may be evaluated through metrics such as job placement rates, role progression, certifications attained, and contributions to significant projects ([Attachment C9-04 – The PEO and its Indicators](#)).

3. Alumni Tracking and Career Progression Analysis

The department utilizes alumni tracking tools to monitor career growth and engagement in professional fields aligned with PEOs. This data is analyzed to assess the effectiveness of the program in preparing graduates for roles in construction, structural engineering, transportation, and water resources, among other specializations. These analyses are compiled in periodic reports that provide

insights into graduates' career paths, skill development, and advancement in their chosen fields, directly contributing to the assessment of PEO 1.

4. Community Engagement and Professional Development

For PEO 2, which emphasizes the graduates' commitment to community and nation-building through involvement in professional organizations, the evaluation includes an assessment of alumni participation in professional and allied organizations, as well as their contributions to community-oriented projects. Regular surveys assess alumni engagement in community development initiatives, leadership roles within professional bodies, and collaborative projects, providing concrete data on the graduates' commitment to social and professional responsibilities.

5. Continuous Learning and Skill Advancement

PEO 3 addresses the graduates' involvement in ongoing education, training, and development within their fields. Evaluation for this objective includes tracking alumni participation in workshops, certifications, advanced degrees, and specialized training programs. Data from alumni surveys and professional development records are analyzed to determine the level of commitment to lifelong learning and skills advancement, ensuring that graduates remain competitive and capable of adapting to industry trends and technological advancements.

6. Data Analysis and Reporting

The CQI Committee aggregates and analyzes data collected through the above methods, comparing it against established benchmarks for each PEO. The results are then synthesized into an evaluation report, detailing the attainment levels of each PEO based on the indicators. The report identifies areas where graduates excel and areas needing improvement, ensuring that each PEO remains a valid reflection of industry standards and professional expectations ([Attachment C9-05 – PEO Result Report](#)).

7. Continuous Improvement and Action Plans

Based on the evaluation findings, the CQI Committee, along with program administrators, develops action plans to address any gaps in achieving the PEOs. This may involve curriculum revisions, enhancement of student support services, introduction of new training programs, or increased collaboration with industry

partners. The CQI cycle ensures that these changes are monitored and reassessed regularly, allowing the department to maintain high educational standards and continually improve alignment with the PEOs ([Attachment C9-06 – Minutes of the Meeting discussing PEO result](#)).

Program Educational Objectives (PEO)

The PEO used in the survey was the PEO revised in 2022 as follows:

The alumni of Bachelor of Science in Civil Engineering (BSCE) three to five years after graduation shall:

PEO 1: Practice in the fields of Construction Engineering and Management, Structural Engineering, Transportation Engineering, or Water Resources Engineering, or Geotechnical Engineering, or other related fields.

PEO 2: Become committed members of professional and other allied organizations engaging in community development and nation-building.

PEO 3: Engage in continuous training and development in current or emerging trends, and advancement in their chosen field of specialization.



These objectives appear in the departmental bulletin board, in the Polytechnic University of the Philippines website (<https://www.pup.edu.ph/ce//bsce>), in the program catalog, and course syllabi.



Figure 9.3 Evaluation Procedure for the PEOs

The Civil Engineering Department at PUP conducts a comprehensive assessment and evaluation of the program's effectiveness on an annual basis. Each year, the Continuous Quality Improvement (CQI) Committee facilitates a systematic review of employer surveys, tracer studies, and stakeholder feedback to determine the extent to which alumni are attaining the Program Educational Objectives (PEOs). The results of these annual assessments are consolidated and presented to the Department for deliberation, after which appropriate actions are identified to ensure continuous quality improvement. The assessment started in 2025, when the 2022 graduates had already reached the three-year post-graduation period. To uphold a high standard of quality, the CQI Committee has established a minimum mean score of 3.0 on a 5-point rating scale or 60% as the benchmark for acceptable PEO attainment.

In addition, the Department follows the PTC-prescribed guideline of assessing PEO attainment three years after the graduation of each batch of students. For instance, under the 2018 curriculum, the first cohort graduated in 2022 and was formally evaluated in 2025. This

year thus marks the initial implementation of the PEO evaluation cycle for graduates of the 2018 curriculum. Moving forward, succeeding cohorts will continue to be evaluated on the same 3-year post-graduation schedule, ensuring that the Department obtains valid evidence of alumni performance in professional practice and uses the findings to strengthen program quality.

Frequency of Assessment and Evaluation

A systematic schedule is implemented for the assessment and evaluation of the Program Educational Objectives (PEOs) to ensure that the program remains relevant, responsive to stakeholder needs, and aligned with the outcomes-based education framework required by the Philippine Technological Council. The assessment and evaluation activities are conducted at regular intervals to provide reliable data for the continuous improvement of the program.

The assessment of Program Educational Objectives is generally conducted three to five years after graduation, which corresponds to the period in which graduates are expected to demonstrate the accomplishments described in the PEOs. This timeframe allows the program to gather meaningful information regarding the professional development, career progression, and contributions of alumni in their respective fields.

Several assessment processes are carried out at specific intervals. Alumni tracer studies and graduate surveys are typically conducted every 3 years after graduation to gather information on employment status, professional achievements, continuing education, and leadership roles of graduates. These surveys provide important data that indicate whether the Program Educational Objectives are being achieved.

Similarly, employer surveys are conducted three to five years after graduation to obtain feedback from organizations that employ the program's graduates. The surveys assess graduates' professional competence, technical capabilities, teamwork, communication skills, ethical responsibility, and leadership potential in the workplace.

In addition to surveys, the program regularly engages with industry and other stakeholders through Government-Industry-Academe Board (GIAB) Committee meetings and stakeholder consultations, which are usually conducted annually or as needed. These meetings provide valuable insights regarding current industry trends, workforce expectations, and emerging technologies that may influence the relevance of the Program Educational Objectives.

The evaluation of the collected assessment data is conducted through faculty meetings, curriculum committee deliberations, and program review sessions, which are typically held once every academic year. During these meetings, the faculty analyzes the assessment

results, determines the level of attainment of the Program Educational Objectives, and identifies areas that require improvement.

The results of these evaluations are documented and used as a basis for continuous quality improvement initiatives, such as curriculum enhancement, integration of new technologies, strengthening of industry partnerships, and refinement of the Program Educational Objectives. Through this regular and systematic schedule of assessment and evaluation, the Polytechnic University of the Philippines ensures that its academic programs remain responsive to stakeholder expectations and maintain high-quality standards.

Satisfactory Level of Attainment for each of the Program's Educational Objectives

Clear performance expectations are established for the attainment of the Program Educational Objectives (PEOs) in alignment with the Continuous Quality Improvement (CQI) framework and the standards of the Philippine Technological Council. To ensure measurable evaluation, each Program Educational Objective is associated with a defined satisfactory level of attainment and supported by documented assessment results.

For each Program Educational Objective (PEO), the achievement of the corresponding Performance Indicators (PIs) by a graduate cohort is evaluated through employer and alumni surveys and/or structured interviews conducted as part of the formal PEO assessment cycle. Alumni are expected to demonstrate attainment of at least one PI for each of the three established PEOs. The Department has set the **expected satisfactory level of attainment at a minimum of 60% of the graduate cohort achieving every PEO**. When this benchmark is met, the PEOs are deemed satisfactorily attained.

Should the measured attainment fall below the benchmarked PI for several consecutive cohorts, a program-level Continuous Quality Improvement (CQI) action is initiated. This may include curriculum revision, enhancement of instruction, or the strengthening of industry-academe linkages to better support PEO achievement. Results of the PEO attainment assessments are systematically documented and consolidated into an annual outcomes assessment report, which includes attainment data, analysis of results, and recommended CQI measures. These reports are presented to the Department, College, and University Academic Council for review and are archived as official evidence of compliance.

For the PEO assessment of graduates under the 2018 curriculum who completed the program in 2022, a total of 135 graduates were identified as the target respondents for the survey ([Attachment C9-07 – List of the 2022 Graduates](#)).

Out of this total, 75 graduates, or 55.55%, responded to the survey. This response rate is considered acceptable, as it exceeds the institutionally approved minimum survey response rate of 30%. In the case of the employer survey, responses were obtained from 68 employers, representing 50.37% of the total graduate population. Although the employer response rate is

slightly lower than that of the graduates, it likewise exceeds the institutional minimum requirement and is therefore considered sufficient to provide meaningful input for the evaluation of Program Educational Objectives (PEOs). Alternatively, the required number of survey respondents may be determined using **Slovin's formula**. Based on a total population of 135 and a 10% margin of error, the **computed minimum sample size is 58 respondents**. The current survey meets this requirement.

Table 9.1 Number of Respondents on the PEO Survey

Respondents	Total Number of Graduates	Number Responded	Percentage
Graduates	135	75	55.55 %
Employers	135	68	50.37%

Number of PEO Respondents

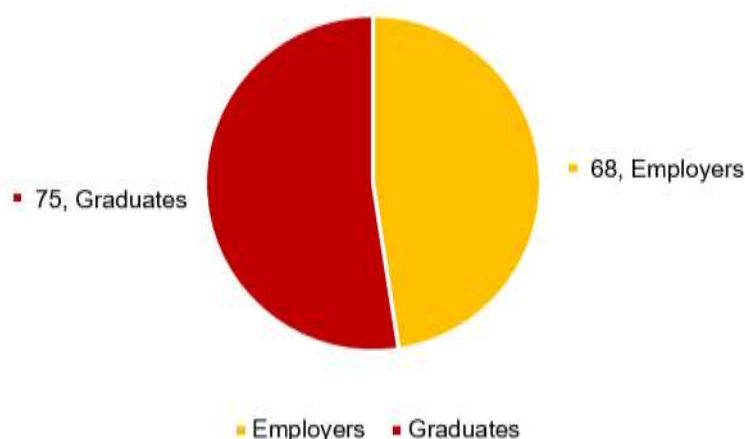


Figure 9.4 Number of Respondents on the PEO Survey

The attainment report is reviewed during faculty meetings, program assessment sessions, and curriculum committee deliberations and is formally documented as part of the program's quality assurance records.

Through this structured approach, PUP ensures that the attainment of Program Educational Objectives is measurable, evidence-based, regularly monitored, and continuously improved, demonstrating compliance with quality standards and commitment to academic excellence.

PEO Assessment Result

A systematic evaluation of the attainment of the Program Educational Objectives (PEOs) is conducted by comparing the actual level of attainment with the established expected

satisfactory level for each objective. This process is aligned with the Continuous Quality Improvement framework and the assessment requirements of the Philippine Technological Council.

Based on the PEO assessment results, both the alumni and employer survey responses indicate that all Program Educational Objectives were attained, since all performance indicators obtained ratings equal to or above the 60% threshold. Overall, the results show a positive validation from both stakeholder groups regarding the effectiveness of the program in achieving its intended educational objectives. ([Attachment C9-08 – Alumni Survey Result](#)) ([Attachment C9-09 – Employer Survey Result](#))

For PEO 1, which pertains to graduates' competence in their chosen field of specialization such as Construction Engineering and Management, Structural Engineering, Transportation Engineering, Water Resources Engineering, Geotechnical Engineering, and other related areas, the results show consistently favorable ratings from both alumni and employers. Alumni ratings ranged from 68.89% to 72.78%, while employer ratings ranged from 70.20% to 71.20%. These figures indicate that graduates have demonstrated the expected technical expertise, management skills, and adherence to national and global requirements, policies, and standards of civil engineering practice. Since all indicators are above the 60% benchmark, PEO 1 is considered attained based on both alumni and employer feedback.

For PEO 2, which focuses on graduates as committed members of professional and allied organizations who engage in community development and nation-building, the results also meet the required threshold, although this objective posted the lowest ratings among the three PEOs. Alumni responses ranged from 61.11% to 67.22%, while employer responses ranged from 60.00% to 62.00%. These results suggest that graduates are participating in professional organizations, contributing to societal interests, and upholding ethical standards in the profession, though the attainment levels are more modest compared with the other objectives. Still, because all values meet or exceed the 60% passing threshold, PEO 2 is likewise considered attained. The results may also indicate an area for continued enhancement, particularly in strengthening professional involvement, civic engagement, and ethical leadership among graduates.

For PEO 3, which addresses graduates' involvement in continuous training and development in current or emerging trends and advancements in their specialization, both alumni and employer responses again confirm attainment. Alumni ratings were 67.78% and 70.56%, while employer ratings were 62.00% and 68.00%. These findings indicate that graduates are pursuing further studies, career advancement, and professional development activities such as trainings and seminars relevant to community needs and current industry trends. Since all indicators exceeded the threshold, PEO 3 is considered attained by both groups of respondents.

Comparing the two respondent groups, the alumni ratings are generally slightly higher than the employer ratings across most indicators. This may suggest that graduates perceive themselves to be performing somewhat more strongly in relation to the PEOs, while employers provide a slightly more conservative but still affirming assessment. Despite these differences, both groups consistently rated all indicators above the required benchmark, thereby reinforcing the conclusion that the program has successfully attained its educational objectives.

In summary, the PEO survey results demonstrate that all three Program Educational Objectives were attained based on both alumni and employer assessments, using the 60% threshold as the basis for attainment. The strongest results were observed under PEO 1, highlighting graduate competence in technical practice and leadership within their field of specialization, while PEO 2 posted the lowest yet still passing results, suggesting a possible area for further improvement in future CQI initiatives. Overall, the findings provide evidence that the program is effective in producing graduates who are technically competent, professionally engaged, and committed to lifelong learning and development.

Although the results of the alumni and employer surveys indicate that all Program Educational Objectives (PEOs) were attained, with all indicators meeting or exceeding the 60% threshold, the department recognizes that attainment alone should not be the endpoint of quality assurance efforts. Rather, these results should serve as a basis for pursuing continuous quality improvement initiatives aimed at further increasing the level of PEO attainment in future assessment cycles. In particular, the department may consider strengthening strategies related to stakeholder engagement, professional involvement, lifelong learning, industry linkage, and curriculum enhancement to address areas with comparatively lower ratings. Possible initiatives may include closer collaboration with employers, expanded alumni tracking and engagement, increased participation in professional organizations, enhancement of extension and community-based activities, and the integration of emerging industry trends into the curriculum and co-curricular programs. Through these proactive measures, the department can build on the current positive results and further improve the extent to which graduates achieve the desired educational objectives.

Table 9.2 Result of PEO Assessment

PROGRAM EDUCATIONAL OBJECTIVES (PEO)	Performance Indicator	Alumni	Employer	Remarks
PEO 1: In the field of Construction Engineering and Management, or Structural Engineering, or Transportation Engineering, or Water Resources Engineering, or Geotechnical Engineering, or other related fields	1. Graduates demonstrate technical expertise within their chosen field of specialization in civil engineering practice	68.89%	71.20%	Attained
	2. Graduates develop management skills to lead within their chosen field of specialization in civil engineering practice	72.78%	70.80%	Attained
	3. Graduates adhere to the national and global requirements, policies and standards of the civil engineering practice	72.22%	70.20%	Attained
PEO 2: Committed members of professional and other allied organizations engaging in community development and nation-building.	1. Graduates are committed members of professional and other allied organizations	61.11%	60.00%	Attained
	2. Graduates are members of organizations adhering to the interests of society.	63.33%	60.00%	Attained
	3. Graduates participate in the upholding of ethical standards in the civil engineering profession.	67.22%	62.00%	Attained
PEO 3: Involved in continuous training and development in current or emerging trends, and advancement in their chosen field of specialization.	1. Graduates are pursuing studies and academic activities for their career advancement in their chosen field of specialization	67.78%	62.00%	Attained
	2. Graduates are involved in training and seminars in addressing community needs, current trends and advancement in their	70.56%	68.00%	Attained

	chosen field of specialization			
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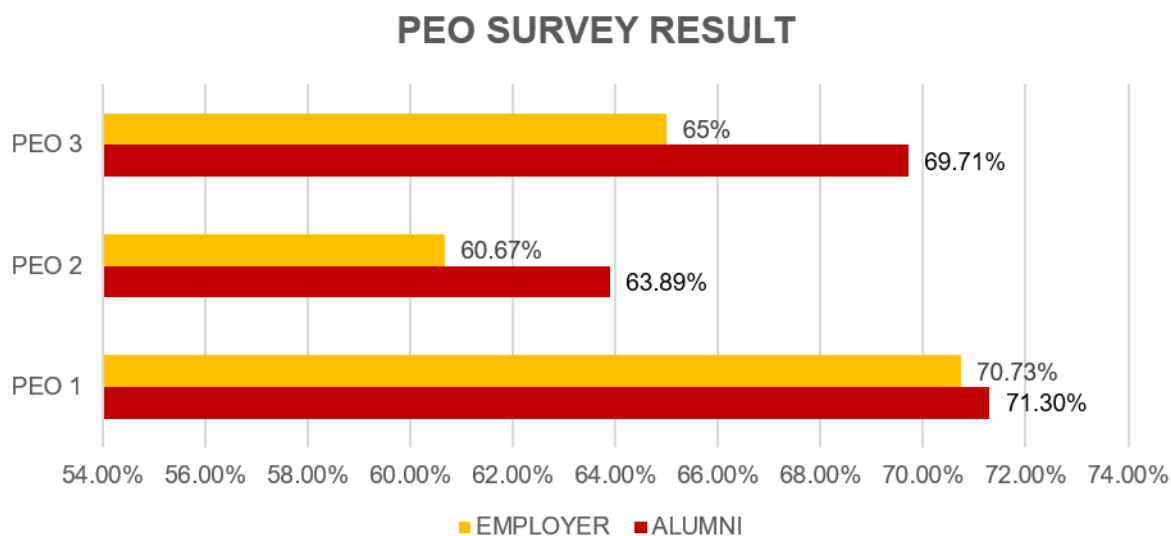


Figure 9.5 Result of PEO Assessment

The results of this comparison are discussed during faculty meetings, program assessment reviews, and curriculum committee deliberations, and are formally documented as part of the program's quality assurance records.

Through this systematic comparison of actual versus expected attainment, PUP ensures transparency, data-driven decision-making, and continuous enhancement of its academic programs in alignment with professional standards and quality assurance requirements.

Graduate Tracer Study

Another assessment that was conducted to assess the Graduates is the Graduate Tracer Study. The survey was completed by graduates of batch 2022, which consisted of graduates from the 2018 curriculum and other curriculum years. No sampling method is used since all the alumni are the target respondents. The study used a redesigned version of the tracer study survey questionnaire utilized from 2017 to 2019. Respondents completed an online survey. The Institute for Data and Statistical Analysis (IDSA) summarized and examined the instrument's data ([Attachment C9-10 – Graduate Tracer Study 2022](#)).

Out of the 205 graduates, 82 or 40% responded to the online graduate tracer survey. Based on the data on current employment, 74 or 90.24% are employed, 2 or 2.44% are self-employed and 3 or 2.66% are unemployed, while 3 or 3.66% has never been employed. With the data gathered, the IDSA had computed that the employment rate is 92.68%.

The table presents the Graduate Tracer Study result for the Bachelor of Science in Civil Engineering program and shows the length of time it took graduates to obtain employment

after graduation. Based on the data, a total of 82 graduates were included in the study. The largest group of graduates, consisting of 30 or 36.59%, were hired within 1 to 3 months after graduation. This is followed by 25 graduates or 30.49% who were able to secure employment in less than 1 month, indicating that a significant number of graduates were employed shortly after completing the program. Meanwhile, 8 graduates or 9.76% found employment within 4 to 6 months, and 11 graduates or 13.41% were hired within 7 to 12 months. A smaller number, 3 graduates or 3.66%, obtained employment within 13 to 18 months after graduation.

Table 9.3 also shows that no graduates were reported to have been hired within 19 to 24 months or more than 2 years, which may suggest that those who eventually became employed were generally able to do so within a shorter period. In addition, 2 graduates or 2.44% were reported as self-employed, reflecting that some graduates pursued independent or entrepreneurial career paths. On the other hand, 3 graduates or 3.66% were identified as never having been employed at the time of the survey.

Overall, the results of the graduate tracer study indicate a favorable employability outcome for graduates of the BS Civil Engineering program. The fact that the majority of graduates were able to secure employment within the first three months after graduation suggests that the program is generally effective in preparing students for the labor market. This reflects positively on the relevance of the curriculum, the adequacy of training and field exposure, and the readiness of graduates to meet industry demands.

The tracer study therefore serves as an important indicator of program effectiveness and provides useful input for continuous quality improvement initiatives of the department.

Table 9.3 Graduate Tracer Study Result

Degree Program	Current Employment	Frequency	Percentage (%)
Bachelor of Science in Civil Engineering	Employed	74	90.24
	Self-employed	2	2.44
	Unemployed	3	3.66
	Neven Been Employed	3	3.66
	EMPLOYMENT RATE	92.68%	
	Time to get Hired after Graduation	Frequency	Percentage (%)
	Less than 1 month	25	30.49
	1 – 3 months	30	36.59
	4 – 6 months	8	9.76
	7 – 12 months	11	13.41
	13 to 18 months	3	3.66
	19 to 24 months	-	-
	More than 2 years	-	-
	Self-employed	2	2.44
	Never been employed	3	3.66
	TOTAL	82	100

Research Studies Conducted on Graduates

Table 9.4 shows the list of faculty research of the Civil Engineering Department. The department undertakes research-based initiatives to assess the efficiency and effectiveness of its curricular activities as part of its overall assessment process. These studies reflect the department's commitment to evaluating different aspects of curriculum implementation, instructional delivery, student preparation, and experiential learning. Through these faculty-led research efforts, the department is able to generate evidence-based findings that help identify strengths, gaps, and areas that may require enhancement. The results of such assessments serve as valuable inputs in planning and implementing Continuous Quality Improvement (CQI) measures, thereby supporting the sustained improvement of the Civil Engineering program and its alignment with desired student outcomes and program objectives ([Attachment C9-11 – Copies of the Research Papers on Assessment](#)).

Table 9.4 List of Faculty Research on Assessment

Title	Authors	Remarks
Surveying the Alignment of Industry Expectation and Perceived Academic Preparation of Civil Engineering Students	Florante D. Poso, Jr., Lorenzo Galicia, Kenneth Bryan M. Tana, Ruffa S. Roxas	Presented in 2025 PAES General Assembly Presented in 2025 PICE Midyear Conference
Assessing the Effectiveness of Civil Engineering OJT Programs: Insights from Students, Industry Partners, and Faculty Supervisors	Florante D. Poso, Jr., Ruffa S. Roxas, Edna P. Arrojado, Ramir M. Cruz, Razon C. Domingo	Presented in 2025 PAES General Assembly
Insights from 350 Students, Industry Partners, and Faculty Supervisors Evaluating The Effectiveness of Civil Engineering O.J.T. Program at P.U.P.	Florante D. Poso, Jr., Ruffa S. Roxas, Edna P. Arrojado, Ramir M. Cruz, Razon C. Domingo, and Kenneth Bryan M. Tana	Presented in 2025 PICE National Conference
AI-Enabled Curriculum Design for Future-Ready Civil Engineering Education: Insights from Lecturers	Florante D. Poso, Jr., Edna P. Arrojado, and Kenneth Bryan M. Tana	Presented in Tokyo, Japan during the 12th International Conference on Frontiers of Educational Technologies (ICFET 2026)
Exploring the Role of Artificial Intelligence in Shaping Learning Outcomes and Student Confidence among Civil Engineering Students	Florante D. Poso, Jr., and Melvin G. Singayan	Presented in Tokyo, Japan during the 12th International Conference on Frontiers of Educational Technologies (ICFET 2026)

Board Examination Results Analysis

Another method of assessing the quality of graduates is the licensure examination results. The summary of the Civil Engineering licensure examination results shows that, over a ten-year period, the PUP Civil Engineering Department consistently surpassed the national average passing rate. This sustained performance indicates that the department has maintained a strong level of effectiveness in preparing its graduates for the licensure examination and for professional engineering practice. The fact that PUP performed above the national passing rate throughout the entire period reflects the quality of its academic instruction, the relevance of its curriculum, and the adequacy of its review and preparation

mechanisms. More importantly, this trend suggests a stable culture of academic excellence and continuous improvement within the department. Consistently exceeding the national passing performance for ten years is a significant indicator of program strength and serves as strong evidence that the Civil Engineering program of PUP is producing graduates who are well-prepared, competitive, and capable of meeting the standards of the profession ([Attachment C9-12 – Licensure Examination Data](#)).

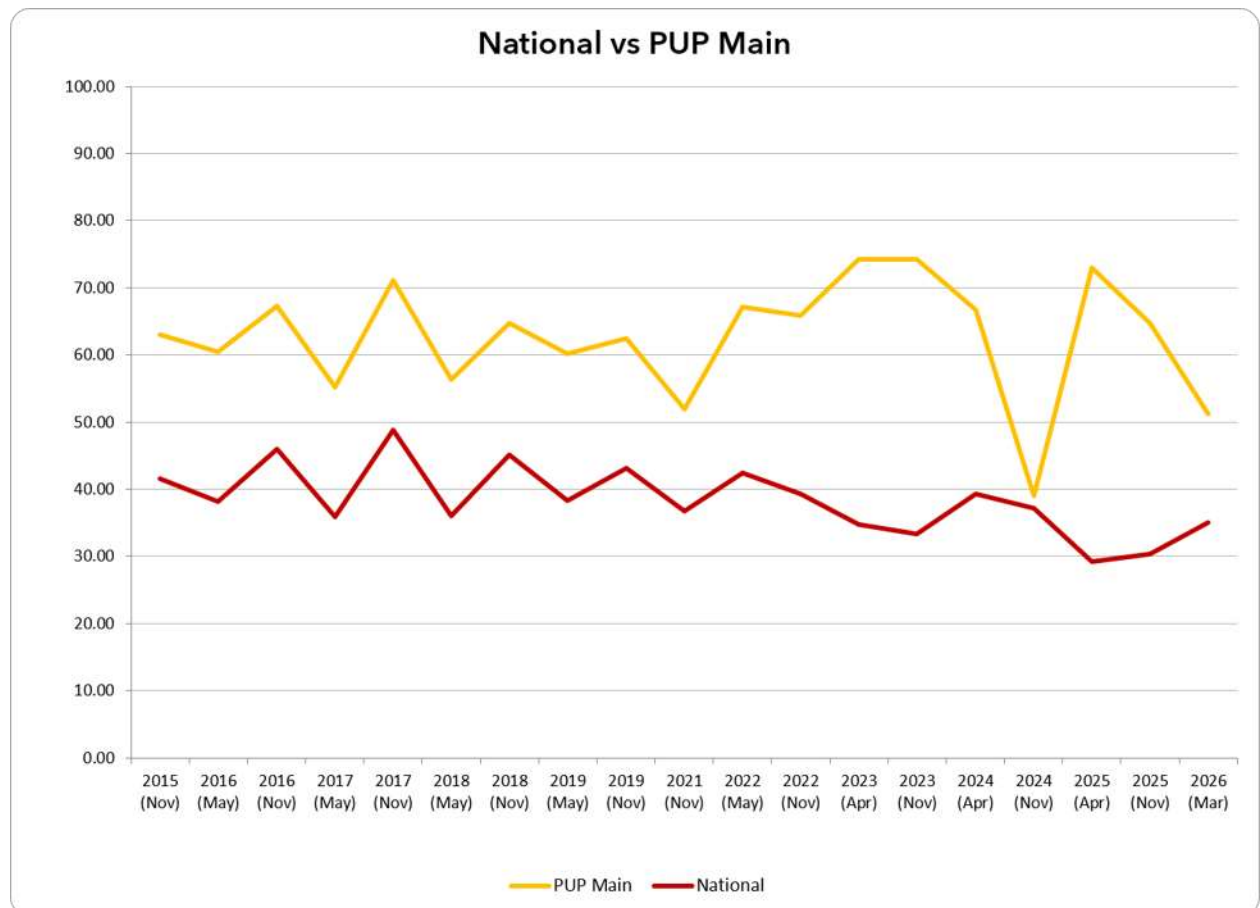


Figure 9.6 Civil Engineering Licensure Examination Results

The Civil Engineering Department at PUP maintains a strong commitment to continuous quality improvement (CQI) to ensure that the Program Educational Objectives (PEOs) and Student Outcomes (SOs) remain relevant and aligned with institutional goals, industry needs, and accreditation standards. Based on the results of outcomes assessment, several continuous improvement actions have been undertaken in recent years. These include the refinement of course syllabi to strengthen the mapping of outcomes, the introduction of enhanced instructional methodologies to address gaps in direct teaching effectiveness, and the integration of industry feedback into curriculum enrichment. Employer and alumni survey results have also been used to calibrate PEO attainment benchmarks and to update program practices in line with emerging professional competencies ([Attachment C9-13 – MOM of Department Meetings](#)).

In addition, the Department has institutionalized regular consultations with the Advisory Board to validate findings and ensure that program improvements are responsive to the needs of employers and other stakeholders. Deliberations from these meetings have resulted in targeted interventions, such as strengthening linkages with industry partners to provide more robust internship opportunities, expanding research and extension activities, and incorporating sustainability principles in core and elective courses. A meeting with the GIAB was conducted on March 16, 2026, to which the PEO survey result was presented and discussed. Another meeting with the faculty members was conducted to scrutinize and discuss the PEO result and identify measures to increase the percentage attainment. ([Attachment C9-14– Department Minutes of the Meeting](#)) ([Attachment C9-15 - MOM of Department's Meeting](#))

Looking ahead, the Department has identified several significant future improvement plans. These include the development of a more comprehensive digital database for storing and tracking assessment results, the expansion of employer and alumni surveys to cover a broader base of respondents, and the enhancement of faculty development initiatives to ensure continuous pedagogical innovation. The Department also plans to align its future CQI cycle more closely with the Philippine Technological Council (PTC) outcomes-based accreditation framework, ensuring systematic monitoring, periodic benchmarking, and timely implementation of responsive actions. These planned initiatives are designed to further elevate the quality of the program and to better prepare graduates for professional engineering practice in an increasingly complex and globalized environment.

The assessment of Program Educational Objectives (PEOs) is conducted through an online survey instrument administered via Google Forms. The survey link is disseminated to alumni and their respective employers through email or various social media platforms to maximize response rates. Once responses are collected, the Civil Engineering Department's Continuous Quality Improvement (CQI) Team undertakes the evaluation and statistical analysis of the data to measure the level of PEO attainment against established benchmarks.

The results of the analysis are consolidated into a formal PEO Assessment Report, which includes statistical summaries, attainment levels, interpretation of findings, and recommended CQI actions. This report is presented during the Department Meeting and subsequently deliberated upon in the Government Industry Advisory Board (GIAB) Meeting to solicit input from industry partners and other stakeholders. The finalized report is then endorsed and submitted to the College Quality Assurance Office and the University's Institutional Quality Assurance Unit for record-keeping, accreditation compliance, and long-term documentation. All PEO assessment reports are securely archived both in digital and printed formats to ensure accessibility for future reference, monitoring, and continuous improvement cycles.

B. Student Outcomes

1. A listing and description of the assessment processes used to gather the data upon which the evaluation of each student outcome is based. Examples of assessment processes may include, but are not limited to, specific exam questions, student portfolios, internally developed assessment exams, senior project presentations, nationally-normed exams, oral exams, focus groups, industrial advisory committee meetings, or other processes that are relevant and appropriate to the program.
2. A listing and description of the evaluation processes for each of the student outcomes.
3. The frequency with which these assessment and evaluation processes are carried out.
4. The expected satisfactory level of attainment for each of the student outcomes. This should include report on the performance indicators for each of the student outcome.
5. Actual level of attainment versus the expected satisfactory level for each student outcome.
6. Continuous improvement actions undertaken and any significant future improvement plans.
7. Documentation and tracking of the results.

The degree of attainment for the student outcomes (SO) is assessed and evaluated by the Civil Engineering program using a well-established, suitably documented process. As students gradually achieve the desired SO, the findings and actions resulting from the assessment processes are used to identify areas where adjustments should be made to improve the quality of institutional learning outcomes further. This allows the program's continuous improvement plans and actions to be implemented based on the degree to which student outcomes are attained. Below is a description of the information gathering, documentation, assessment, and evaluation processes for student outcomes.

Due to the stipulations of Republic Act 10533, referred to as the "Enhanced Basic Education Act of 2013," which includes guidelines for curriculum mapping in terms of Introduce–Reinforce–Demonstrate (I/R/D), K–12 programs began to be implemented in the academic year 2018–2019 ([Attachment C9-16 – RA 10533](#)).

A systematic and structured process is implemented for assessing and evaluating the attainment of Student Outcomes (SOs) as part of the Continuous Quality Improvement (CQI) framework. This process is aligned with the outcomes-based education requirements of the Philippine Technological Council and ensures that students demonstrate the knowledge, skills, professional competencies, and attributes expected upon graduation.

Student Outcomes represent the measurable competencies that students must achieve through the curriculum. The program regularly gathers and analyzes assessment data to determine the extent to which these outcomes are attained.

Assessment Methods

The program utilizes multiple and complementary assessment methods to ensure a comprehensive evaluation of student performance. These assessment processes are systematically implemented and documented to support evidence-based improvement.

There are two methods of assessment used in measuring the Student Outcomes:

1. Direct Assessment
2. Indirect Assessment

The first assessment method used to evaluate student outcomes is the direct method using rubrics. The following rubrics are used by the faculty in the civil engineering department to evaluate the level of attainment of SO: ([Attachment C9-17 – Rubric Forms](#))

1. Rubrics for Oral Defense
2. Rubrics for Thesis Proposal Document
3. Rubrics for Final Thesis Document
4. Rubrics for the Performance of the Laboratory Activity
5. Rubrics for the Laboratory Activity Report
6. Rubrics for the Computational Template
7. Analytic Rubric for Complexity of Engineering Activities
8. Analytic Rubric for Complexity of Engineering Problems
9. Complexity Analysis Matrix of Complex Engineering Problem
10. Complexity Analysis Matrix of Complex Engineering Activity

OJT, SEMINAR, AND CE PROJECT

The assessment of Student Outcomes (SOs) in the Civil Engineering program is carried out using both (1) direct and (2) indirect assessment methods to ensure a comprehensive evaluation of student performance and learning achievement. The direct assessment method primarily uses developed rubrics to measure the extent to which students demonstrate the

expected outcomes through actual performance in various curricular activities and academic outputs. These rubrics are systematically applied by the faculty members of the Civil Engineering Department in evaluating student work in key courses and learning experiences. Among the rubrics used are the Rubric for Oral Defense, Rubric for Thesis Proposal Document, Rubric for Final Thesis Document, Rubric for the Performance of the Laboratory Activity, Rubric for the Laboratory Activity Report, and Rubric for the Computational Template. These assessment tools enable the department to gather measurable and outcome-based evidence on students' knowledge, technical competence, communication skills, problem-solving ability, and other competencies embodied in the Student Outcomes. Since the rubrics are aligned with the performance indicators of the SOs, the results derived from these tools provide reliable input in determining the level of attainment of each outcome.

In addition to direct assessment, the department also employs an indirect assessment method through student surveys on the attainment of Student Outcomes. In this process, students are asked to evaluate the extent to which they believe the program has enabled them to develop the competencies reflected in the SOs. This method provides valuable perception-based feedback from the learners regarding their academic experiences and the effectiveness of the teaching-learning process. While indirect assessment does not measure performance in the same way as direct assessment, it serves as an important complementary source of evidence that helps validate and enrich the findings obtained from rubric-based evaluations ([Attachment C9-18 – Sample of Indirect Assessment Survey Form](#)).

By using both direct and indirect methods, the department can establish a more balanced and systematic assessment process for Student Outcomes. The direct method provides objective evidence of student performance, while the indirect method captures students' perceptions and experiences. Together, these assessment approaches support the department's continuous quality improvement efforts by providing meaningful data for evaluating curriculum effectiveness, instructional delivery, and the overall achievement of the program's intended learning outcomes.

SO Assessment Process

Figure 9.7 below shows the assessment process used by the department in assessing the student outcomes.

The assessment process of Student Outcomes in the Civil Engineering Department of PUP begins with the use of summative activities that serve as the primary sources of direct evidence of student learning. These summative activities include major examinations, laboratory practicals and reports, design problems, projects, and other culminating course requirements that are intentionally aligned with specific Student Outcomes and their

corresponding performance indicators. After students complete these activities, faculty members proceed to checking and recording by evaluating outputs using agreed scoring guides or rubrics to ensure that performance is judged against clearly defined criteria. The results are then documented systematically, converting student work into outcome-based data that can be summarized and analyzed across classes and academic terms. Finally, the process emphasizes adherence to measurement by ensuring that all assessment practices follow the department's established measurement plan, including consistent use of rubrics, scoring scales, attainment targets, and documentation procedures. This final step confirms that the collected results are valid, comparable, and reliable for reporting purposes and for guiding continuous improvement efforts in the program.

Through the integration of direct assessments, indirect feedback mechanisms, performance-based evaluations, and stakeholder consultations, PUP ensures a comprehensive evaluation of Student Outcomes. The systematic collection and analysis of assessment data support evidence-based decision-making and continuous improvement of the curriculum.

This structured approach demonstrates compliance with the outcomes-based accreditation framework and reinforces the university's commitment to academic excellence and quality assurance through measurable student outcome attainment



Figure 9.7 The Assessment Process

1. Conduct of Summative Activity

The cornerstone of our student outcome assessment process is the administration of summative class activities. These assessments are carefully designed to directly align with the specific learning outcomes outlined in the curriculum. By employing a diverse range of assessment methods, including examinations, projects, presentations, and research papers, the program ensures a comprehensive evaluation of student learning. These activities are strategically scheduled throughout the semester to provide consistent opportunities for students to demonstrate their mastery of the subject matter.

2. Checking and Recording

Following the completion of each summative class activity, a process of checking and recording student performance commences. Faculty members dedicate significant time to carefully review and grade each assessment, providing detailed feedback to students on their strengths and areas for improvement. The grades and feedback are then meticulously recorded and announced to the students and encoded in the grading system, allowing for easy access and analysis. This systematic approach ensures the accurate and efficient documentation of student achievement.

3. Adherence Measurement

To maintain the integrity of the assessment process and ensure its effectiveness in measuring student learning, the department employs a system of adherence measurement. This involves regular monitoring and evaluation of faculty members' adherence to the established assessment guidelines and rubrics. By conducting periodic reviews of graded assessments, they can identify any inconsistencies or deviations from the expected standards. Through this comprehensive approach to student outcome assessment, the department is committed to providing a rigorous and equitable evaluation of student learning.

The Student Outcomes

The Polytechnic University of the Philippines Civil Engineering department adopts the student outcomes required by the Commission on Higher Education (CHED) and of the Philippine Technological Council (PTC). They are as follows:

SO1. Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex civil engineering problems.

- SO2.** Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
- SO3.** Design solutions for complex civil engineering problems and design systems, components, or processes to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, cultural, and sustainability, in accordance with standards
- SO4.** Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
- SO5.** Identify, formulate, research literature and analyze complex civil engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
- SO6.** Apply ethical principles and commit to professional ethics, and responsibilities and norms of engineering practice.
- SO7.** Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- SO8.** Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a global, economic, societal and environmental context.
- SO9.** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change
- SO10.** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex civil engineering problems.
- SO11.** Create, select and apply appropriate techniques, skills, resources, and modern engineering and IT tools, including prediction and modelling, to complex civil engineering problems with an understanding of the limitations.
- SO12.** Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

These student outcomes are also published in the course syllabi of the students and are aligned to both the course learning outcomes and the program educational objectives ([Attachment C9-19 – Sample Course Syllabi](#)).

For courses that are identified for assessment, the Student Outcome Evaluation Summary Report Form is further summarized and evaluated at the department level for possible improvement as they are directly demonstrative of the student outcomes. Table 9.5 presents the identified courses for assessment for each of the respective student outcomes.

Each Student Outcome is evaluated using measurable performance indicators derived from direct and indirect assessment methods. The evaluation results are analyzed to identify gaps and guide program improvements.

Table 9.5 Identified Courses for Assessment

SN	Student Outcomes	Identified Courses for Assessment
SO 1	Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex civil engineering problems.	Reinforced Concrete Design
SO 2	Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	Civil Engineering Research
SO 3	Design solutions for complex civil engineering problems and design systems, components, or processes to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety,	Civil Engineering Project 2

	manufacturability, cultural, and sustainability, in accordance with standards	
SO 4	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.	On-The-Job Training 2
SO 5	Identify, formulate, research literature and analyze complex civil engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.	Civil Engineering Project 2
SO 6	Apply ethical principles and commit to professional ethics, and responsibilities and norms of engineering practice.	On-The-Job Training 2
SO 7	Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	Civil Engineering Project 1
SO 8	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a global, economic, societal, and environmental context.	Civil Engineering Project 2
SO 9	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change	Civil Engineering Fieldtrip
SO 10	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex civil engineering problems.	Civil Engineering Laws, Ethics and Contracts

SO 11	Create, select, and apply appropriate techniques, skills, resources, and modern engineering and IT tools, including prediction and modeling, to complex Civil Engineering problems with an understanding of the limitations.	Reinforced Concrete Design
SO 12	Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	On-The-Job Training 1 Civil Engineering Project 1

Frequency of the Process

The assessment and evaluation process for the identified courses is conducted based on the scheduled term for data collection, as outlined in Table 9.6. The courses selected for assessment were systematically evaluated across multiple academic terms, including the 1st Semester 2022-2023, 2nd Semester 2022-2023, Summer Term 2022-2023, 1st Semester 2021-2022, and 2nd Semester 2021-2022. This multi-term approach ensures a comprehensive and continuous collection of data, enabling a vigorous evaluation of student outcomes and supporting a data-driven Continuous Quality Improvement (CQI) process.

Table 9.6 Plan for the Direct Assessment of Student Outcomes

Student Outcomes		Performance Indicators		Key Courses	Assessment Methods	Target	Time and Data Collection	Responsibility
SO1	Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex civil engineering problems.	1	Application of knowledge of mathematical models or equations to solve civil engineering problems.	Reinforced Concrete Design	Locally developed exam	60% of the students shall have a rating of at least 50%	4th Year, 1st Semester 2021-2022	Noli Sibayan
		2	Application of knowledge of natural and engineering science to solve complex civil engineering problems.	Reinforced Concrete Design	Locally developed exam	60% of the students shall have a rating of at least 50%	4th Year, 1st Semester 2021-2022	Noli Sibayan
		3	Application of knowledge of engineering fundamentals to the solution of complex	Reinforced Concrete Design	Locally developed exam	60% of the students shall have a rating of	4th Year, 1st Semester 2021-2022	Noli Sibayan

Table 9.6 Plan for the Direct Assessment of Student Outcomes

Student Outcomes		Performance Indicators		Key Courses	Assessment Methods	Target	Time and Data Collection	Responsibility
			civil engineering problems.			at least 50%		
SO2	Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	1	Conduct laboratory experiments or fieldwork in accordance with manuals and civil engineering standards while maintaining orderliness.	Civil Engineering Research	CE Research with Rubrics for Research Paper	60% of the students shall have a rating of at least 50%	3rd Year, 1st Semester 2020-2021	Donamel Saiyari
		2	Design the experimental plan of a laboratory experiment or fieldwork.	Civil Engineering Research	CE Research with Rubrics for Research Paper	60% of the students shall have a rating of at least 50%	3rd Year, 1st Semester 2020-2021	Donamel Saiyari

Table 9.6 Plan for the Direct Assessment of Student Outcomes

Student Outcomes		Performance Indicators		Key Courses	Assessment Methods	Target	Time and Data Collection	Responsibility
		3	Analyze the collected data from laboratory experiment and field works using appropriate methods in accordance to civil engineering principles.	Civil Engineering Research	CE Research with Rubrics for Research Paper	60% of the students shall have a rating of at least 50%	3rd Year, 1st Semester 2020-2021	Donamel Saiyari
		4	Interpret the collected data from laboratory experiment and field works and draw out conclusions using appropriate methods in accordance to civil engineering principles.	Civil Engineering Research	CE Research with Rubrics for Research Paper	60% of the students shall have a rating of at least 50%	3rd Year, 1st Semester 2020-2021	Donamel Saiyari

Table 9.6 Plan for the Direct Assessment of Student Outcomes

Student Outcomes		Performance Indicators		Key Courses	Assessment Methods	Target	Time and Data Collection	Responsibility
SO3	Design solutions for complex civil engineering problems and design systems, components, or processes to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, cultural, and sustainability, in accordance with standards	1	Formulate criteria evaluation and analyze realistic constraints of a design problem.	Civil Engineering Project 2	CE Project 2 with Rubrics for Research Paper	60% of the students shall have a rating of at least 50%	4th Year, 2nd Semester 2021-2022	Ruffa Roxas
		2	Create a design project/ prototype/ system/ component/ process that meets the relevant design codes and standards.	Civil Engineering Project 2	CE Project 2 with Rubrics for Research Paper	60% of the students shall have a rating of at least 50%	4th Year, 2nd Semester 2021-2022	Ruffa Roxas
SO4	Function effectively as an individual and as a member or leader in diverse teams and in	1	Participate, express ideas freely and execute member tasks towards	On-The-Job Training 2	OJT Portfolio Performance Evaluation Rubrics	60% of the students shall	3rd Year, Summer 2020-2021	Nhel Adrian Bartoline

Table 9.6 Plan for the Direct Assessment of Student Outcomes

Student Outcomes		Performance Indicators		Key Courses	Assessment Methods	Target	Time and Data Collection	Responsibility
	multi-disciplinary settings.		attainment of the objectives (idea generation.)			have a rating of at least 50%		
		2	Integrate inputs from team members and make decisions based on objective criteria (decision making.)	On-The-Job Training 2	OJT Portfolio Performance Evaluation Rubrics	60% of the students shall have a rating of at least 50%	3rd Year, Summer 2020-2021	Nhel Adrian Bartoline
SO5	Identify, formulate, research literature and analyze complex civil engineering problems reaching substantiated conclusions using first principles of mathematics, natural	1	Identify complex civil engineering problems and identify the key issues, factors, and concepts affecting the problem.	Civil Engineering Project 2	CE Project 2 with Rubrics for Research Paper	60% of the students shall have a rating of at least 50%	4th Year, 2nd Semester 2021-2022	Edna Arrojado

Table 9.6 Plan for the Direct Assessment of Student Outcomes

Student Outcomes		Performance Indicators		Key Courses	Assessment Methods	Target	Time and Data Collection	Responsibility
	sciences and engineering sciences.	2	Analyze and formulate solutions to complex civil engineering problems.	Civil Engineering Project 2	CE Project 2 with Rubrics for Research Paper	60% of the students shall have a rating of at least 50%	4th Year, 2nd Semester 2021-2022	Edna Arrojado
SO6	Apply ethical principles and commit to professional ethics, and responsibilities and norms of engineering practice.	1	Demonstrate ethical and professional behavior during practice.	On-The-Job Training 2	OJT Portfolio Performance Evaluation Rubrics	60% of the students shall have a rating of at least 50%	3rd Year, Summer 2020-2021	Kenneth Bryan Tana
		2	Demonstrate professionalism by complying to relevant codes and standards.	On-The-Job Training 2	OJT Portfolio Performance Evaluation Rubrics	60% of the students shall have a rating of	3rd Year, Summer 2020-2021	Kenneth Bryan Tana

Table 9.6 Plan for the Direct Assessment of Student Outcomes

Student Outcomes		Performance Indicators		Key Courses	Assessment Methods	Target	Time and Data Collection	Responsibility
						at least 50%		
SO7	Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	1	Use effective written communication by writing grammatically correct and logical reports using appropriate graphics and technical style.	Civil Engineering Project 1	CE Project 2 with Rubrics for Research Paper	60% of the students shall have a rating of at least 50%	4th Year, 1st Semester 2021-2022	Prospero Cabornay

Table 9.6 Plan for the Direct Assessment of Student Outcomes

Student Outcomes		Performance Indicators		Key Courses	Assessment Methods	Target	Time and Data Collection	Responsibility
		2	Use effective oral communication by presenting and defending ideas with sound reasoning and in an articulate manner.	Civil Engineering Project 1	CE Project 2 with Rubrics for Oral Defense	60% of the students shall have a rating of at least 50%	4th Year, 1st Semester 2021-2022	Prospero Cabornay
SO8	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a global, economic, societal, and environmental context.	1	Explain the impacts of civil engineering solutions in a global, economic, environmental, and societal context.	Civil Engineering Project 2	CE Project 2 with Rubrics for Research Paper	60% of the students shall have a rating of at least 50%	4th Year, 2nd Semester 2021-2022	Edna Arrojado
		2	Establish the significance of the solutions in terms of beneficiaries.	Civil Engineering Project 2	CE Project 2 with Rubrics for Research Paper	60% of the students shall have a rating of	4th Year, 2nd Semester 2021-2022	Edna Arrojado

Table 9.6 Plan for the Direct Assessment of Student Outcomes

Student Outcomes		Performance Indicators		Key Courses	Assessment Methods	Target	Time and Data Collection	Responsibility
						at least 50%		
SO9	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change	1	The student can demonstrate self-directed learning through acquired new knowledge from various modes and strategies (research, attendance in conferences and exhibits, site visits.)	Civil Engineering Fieldtrip	Rubrics for the Assessment of Student Reports on Site Inspection and Rubrics for the Assessment of Student Reports on Seminars	60% of the students shall have a rating of at least 50%	4th Year, 2nd Semester 2021-2022	Ramir Cruz
SO10	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and	1	Identify contemporary issues in a chosen civil engineering field of specialization.	Civil Engineering Laws, Ethics and Contracts	Locally developed exam	60% of the students shall have a rating of at least 50%	3rd Year, 2nd Semester 2020-2021	Razon Domingo
		2	Determine effect of contemporary issues	Civil Engineering	Locally developed exam	60% of the	3rd Year, 2nd	Razon Domingo

Table 9.6 Plan for the Direct Assessment of Student Outcomes

Student Outcomes		Performance Indicators		Key Courses	Assessment Methods	Target	Time and Data Collection	Responsibility
	solutions to complex civil engineering problems.		in the civil engineering discipline.	Laws, Ethics and Contracts		students shall have a rating of at least 50%	Semester 2020-2021	
SO11	Create, select, and apply appropriate techniques, skills, resources, and modern engineering and IT tools, including prediction and modeling, to complex civil engineering problems with an understanding of the limitations; and	1	Creation of most appropriate techniques or modern tools for a specific civil engineering task	Reinforced Concrete Design	Rubric Assessment for Design	60% of the students shall have a rating of at least 50%	4th Year, 1st Semester 2021-2022	Noli Sibayan
		2	Selection of skills relevant to the civil engineering practice using modern engineering tools and techniques	Reinforced Concrete Design	Rubric Assessment for Design	60% of the students shall have a rating of at least 50%	4th Year, 1st Semester 2021-2022	Noli Sibayan

Table 9.6 Plan for the Direct Assessment of Student Outcomes

Student Outcomes		Performance Indicators		Key Courses	Assessment Methods	Target	Time and Data Collection	Responsibility
		3	Application of the techniques, resources, skills, and modern engineering tools to address complex civil engineering problem	Reinforced Concrete Design	Rubric Assessment for Design	60% of the students shall have a rating of at least 50%	4th Year, 1st Semester 2021-2022	Noli Sibayan
SO12	Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	1	Use appropriate tools and theories to manage multidisciplinary projects in a team	On-The-Job Training 1	OJT Portfolio Performance Evaluation Rubrics	60% of the students shall have a rating of at least 50%	2nd Year, Summer 2020-2021	Razon Domingo

Table 9.6 Plan for the Direct Assessment of Student Outcomes

Student Outcomes		Performance Indicators		Key Courses	Assessment Methods	Target	Time and Data Collection	Responsibility
		2	Identify and devise solutions to real world problems as a member/leader of a cross-functional team.	Civil Engineering Project 2	CE Project 2 with Rubrics for Research Paper	60% of the students shall have a rating of at least 50%	4th Year, 1st Semester 2021-2022	Edna Arrojado

Satisfactory Level of Attainment for each of the Student Outcomes.

The Civil Engineering Department has established a baseline performance standard, requiring at least 60% of students in each course to achieve a satisfactory level for all student outcomes. This benchmark is subject to annual review and adjustment based on the assessment results. The department aims to progressively raise this standard over time as the program matures and student performance improves. By continuously monitoring and analyzing assessment data, the College can identify areas for improvement and implement targeted interventions to enhance student learning and outcomes.

SUMMARY OF RESULTS

Results of Direct Assessment for SO 1

Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

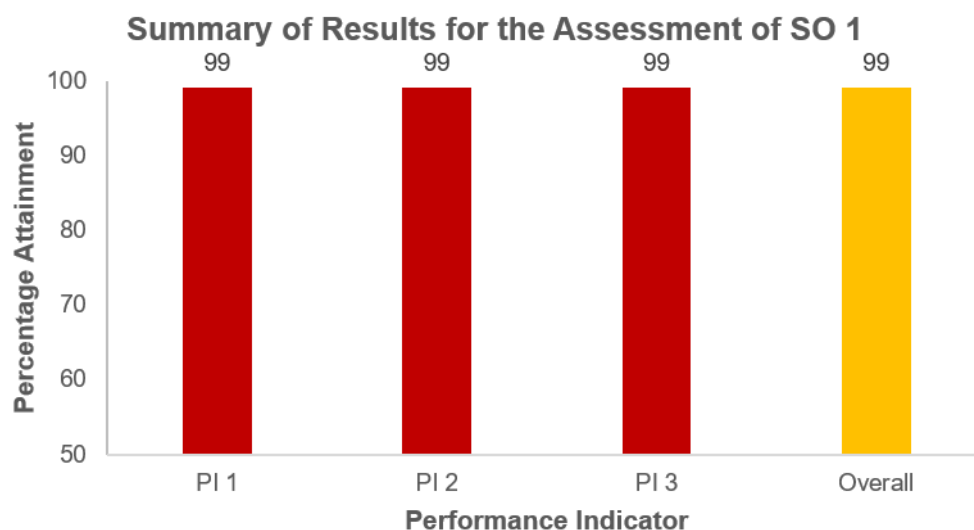


Figure 9.8 SO 1 Summary Result

For the First Semester of Academic Year 2021–2022, the assessment of Student Outcome 1 (SO 1) in the Reinforced Concrete Design subject was conducted across all sections, involving a total of 142 students. The data were collected through a locally developed examination and the final grade, specifically designed to align with the course's intended learning outcomes and the performance indicators under SO 1. These indicators include PI-1 (application of mathematical models or equations to solve engineering problems), PI-2 (application of knowledge of natural and engineering sciences to address complex civil engineering problems), and PI-3 (application of engineering fundamentals in

problem-solving scenarios). The assessment instrument underwent internal validation to ensure that it accurately reflected board-exam level competencies and course coverage.

Results showed that the target proficiency levels were attained across all three performance indicators, indicating satisfactory to exemplary performance. The overall attainment level reached 99%, confirming that Program Outcome 1 was successfully achieved during the period of data collection. It is worth noting that this high performance may have been influenced by the design nature of the course, which is strategically positioned to reinforce previously learned concepts in mathematics, physics, and engineering science. Additionally, heightened student motivation, driven by the proximity to licensure preparation, may have positively impacted performance. However, variations in instructor delivery and students' prior foundational knowledge across sections could be considered as potential factors that may have affected individual results. Overall, the assessment process provided a reliable measure of the students' ability to apply mathematics and science in solving complex civil engineering problems ([Attachment C9-20 – SO 1 Assessment Result](#)).

Results of Direct Assessment for SO 2

Conduct investigations of complex civil engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

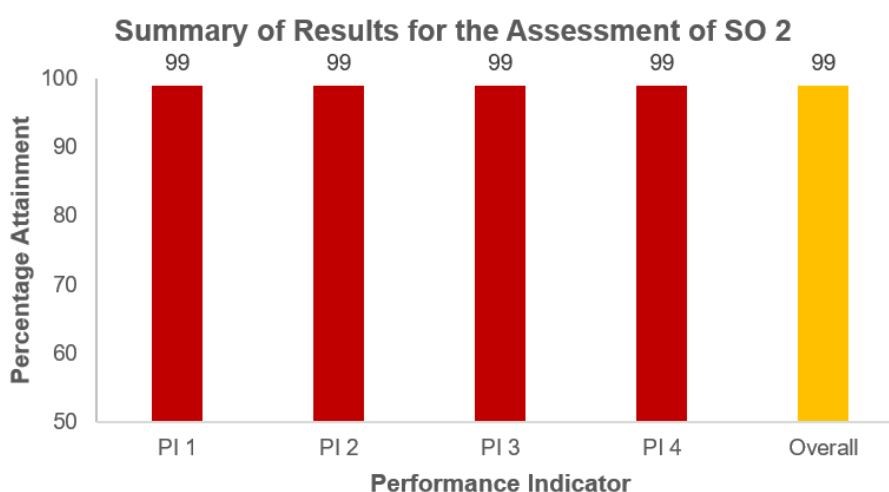


Figure 9.9 SO 2 Summary Result

For the 1st Semester of SY 2021–2022, data were gathered from a total of 212 Civil Engineering students enrolled across all sections in the Civil Engineering Research subject. The performance of students was evaluated using a standardized rubric designed specifically

for assessing research papers. These rubric measured competencies aligned with Program Outcome 2, which focuses on conducting and interpreting experiments and fieldwork in accordance with engineering standards.

Results indicated that the target performance level was attained in all four performance indicators: PI-1 (Conduct laboratory experiment or fieldwork in accordance with manuals and engineering standards while maintaining orderliness), PI-2 (Design the experimental plan of a laboratory experiment or fieldwork), PI-3 (Analyze the collected data from laboratory experiment and field works using appropriate methods in accordance to engineering principles), and PI-4 (Interpret the collected data from laboratory experiment and field works and draw out conclusions using appropriate methods in accordance to engineering principles). Many respondents received scores in the 4–5 range across all indicators, suggesting a strong mastery of the intended outcomes.

It is worth noting that data collection took place during a period of gradual transition back to face-to-face academic activities, which may have positively influenced student performance due to increased collaboration, hands-on engagement, and access to physical resources. Despite potential variations in research topics and methodologies, the consistency of high ratings across indicators reflects the effectiveness of the instructional strategies and assessment tools employed. Overall, an impressive 99% attainment of Program Outcome 2 was recorded, affirming that the course objectives for Civil Engineering Research were successfully met during this term ([Attachment C9-21 – SO 2 Assessment Result](#)).

Results of Direct Assessment for SO 3

Design solutions for complex civil engineering problems and design systems, components, or processes to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, cultural, and sustainability, in accordance with standards

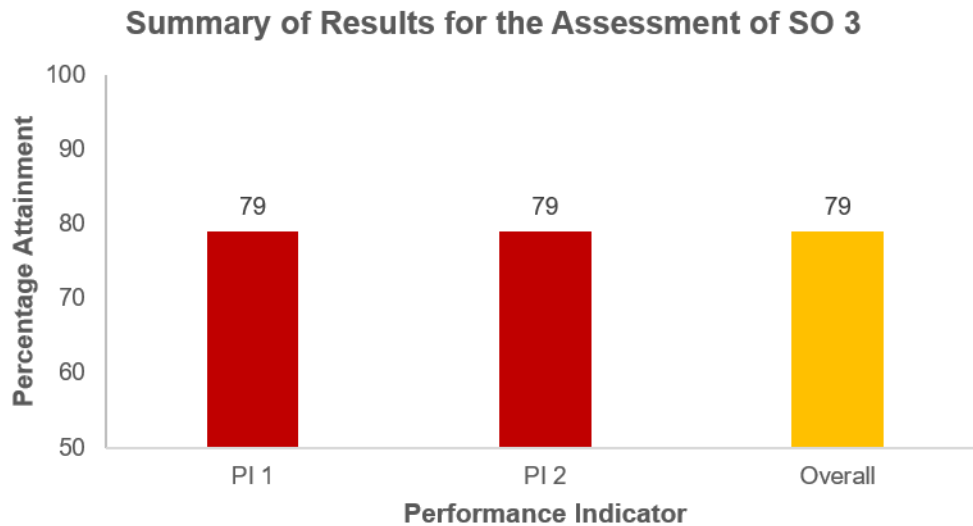


Figure 9.10 SO 3 Summary Result

For the 2nd Semester of SY 2021–2022, a total of 185 students across all sections enrolled in the Civil Engineering Project 2 subject were assessed to determine the attainment of Student Outcome (SO) 3. The data was gathered through the evaluation of final design project outputs and oral presentations using standardized rubrics aligned with the performance indicators: PI-1 (Formulate criteria evaluation and analyze realistic constraints of a design problem) and PI-2 (Create a design project/prototype/system/component/process that meets relevant design codes and standards). The assessment was conducted by a panel of faculty members to ensure consistency and objectivity. Majority of the students received ratings of 4 to 5 across all performance indicators, contributing to an overall attainment rating of 79%, indicating that SO 3 was successfully achieved. It is worth noting that variations in project complexity, group dynamics, and the availability of resources may have influenced student performance. Additionally, the hybrid modality of instruction during the semester, combining limited face-to-face mentoring with online consultations, may have had both enabling and constraining effects on project execution and documentation quality ([Attachment C9-22 – SO 3 Assessment Result](#)).

Results of Direct Assessment for SO 4

Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.

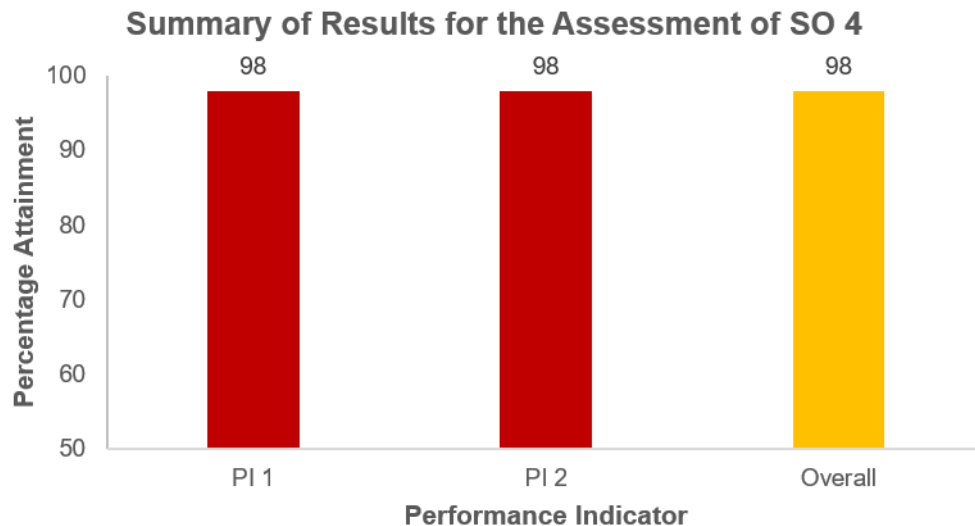


Figure 9.11 SO 4 Summary Result

For the Summer Term of SY 2020–2021, a total of 199 students across all sections were assessed under the On-the-Job Training 2 (OJT 2) course. The primary basis for the assessment was the students submitted OJT portfolios, which included documented activities, daily and weekly logs, accomplishment reports, supervisor evaluations, and self-assessment forms. These portfolios were evaluated by assigned faculty-in-charge using rubrics aligned with the performance indicators (PIs) for Program Outcome 4. Specifically, the assessment focused on PI-1 (participate, express ideas freely, and execute member tasks towards attainment of the objectives) and PI-2 (integrate inputs from team members and make decisions based on objective criteria).

The results showed that a significant majority of students received ratings between 4 and 5 in both indicators, leading to an overall attainment rate of 98% for the specified program outcome. However, certain scenarios may have influenced these results. For instance, the diversity of training environments, ranging from government agencies and private firms to remote or hybrid internship setups, may have affected the depth and nature of team interaction and decision-making tasks. In addition, some companies provided more structured mentoring and feedback mechanisms than others, which could have contributed to the variation in student performance. Nonetheless, the consistency in high ratings across submissions suggests that students were generally able to demonstrate the expected competencies during their respective placements.

The high attainment also reflects the effectiveness of pre-deployment orientations, regular faculty monitoring, and close coordination with host institutions in guiding students throughout their training. These factors are crucial in ensuring that even with differences in

work assignments or settings, the intended learning outcomes of OJT 2 are met ([Attachment C9-23 – SO 4 Assessment Result](#)).

Results of Direct Assessment for SO 5

Identify, formulate, research literature and analyze complex civil engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

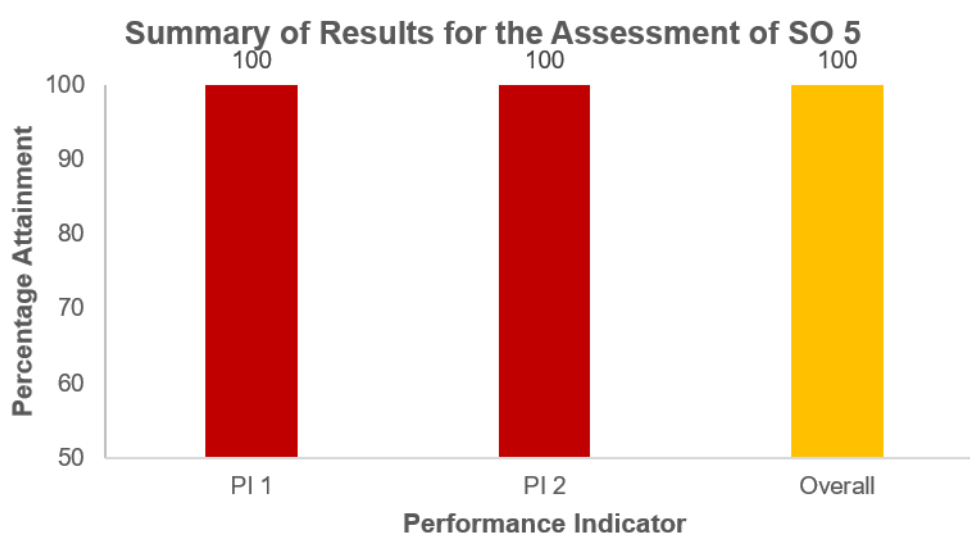


Figure 9.12 SO 5 Summary Result

For the 2nd Semester of SY 2021–2022, assessment data was collected from a total of 143 students enrolled across all sections in the Civil Engineering Project 2 subject. The data was gathered through the evaluation of student outputs, project reports, presentations, and other course-related deliverables that directly align with the targeted performance indicators (PIs). These were measured using a standardized rubric, where students were rated on a scale from 1 to 5, with higher scores indicating higher levels of proficiency and outcome attainment.

Results revealed that the target percentage was successfully achieved in both PI-1 (Identify complex civil engineering problems and identify the key issues, factors, and concepts affecting the problem) and PI-2 (Analyze and formulate solutions to complex civil engineering problems). A significant majority of the students received ratings of 4 or 5 across these performance indicators, which reflects their strong analytical and problem-solving capabilities as expected under Student Outcome 5. This culminated in an overall attainment rating of 100%, signifying full attainment of the desired outcome during the period.

Several factors may have positively influenced the data. First, students had already been exposed to a series of design, theory, and technical courses prior to undertaking Civil Engineering Project 2, which likely strengthened their capacity to handle complex civil engineering problems. Additionally, the collaborative project-based approach of the subject may have encouraged deeper engagement, research, and application of engineering principles.

However, it is also important to note that the implementation of hybrid or flexible learning modes due to the ongoing pandemic may have influenced the learning environment. While some students thrived in independent learning setups, others may have faced challenges related to limited access to resources, internet connectivity, or face-to-face collaboration. Despite these conditions, the data strongly suggests that students were able to meet the course and outcome expectations effectively.

In summary, the high performance across both PIs under Student Outcome 5 demonstrates the effectiveness of the course design and instruction, as well as the students' preparedness and adaptability in addressing complex civil engineering problems ([Attachment C9-24 – SO 5 Assessment Result](#)).

Results of Direct Assessment for SO 6

Apply ethical principles and commit to professional ethics, and responsibilities and norms of engineering practice.

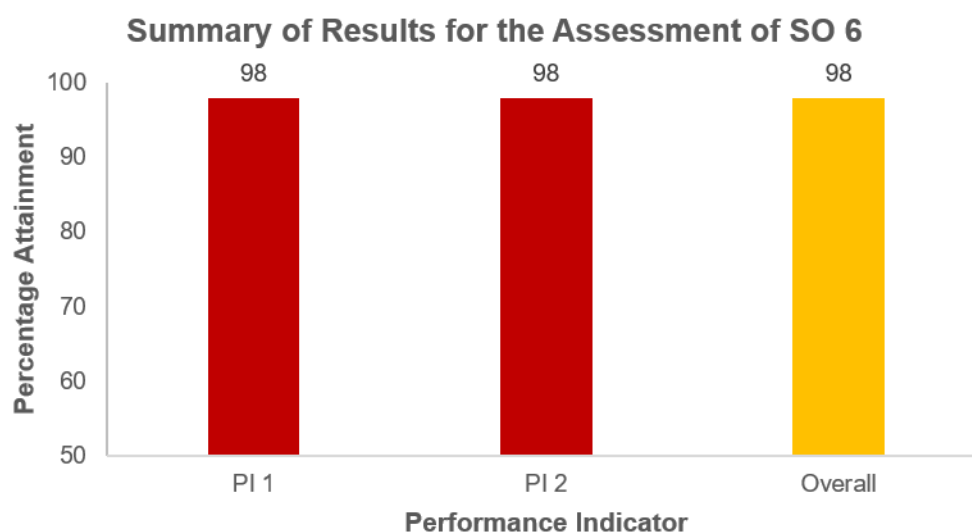


Figure 9.13 SO 6 Summary Result

For the Summer Term of SY 2020–2021, a total of 199 students across all sections were enrolled in the On-the-Job Training 2 (OJT2) course. The assessment of Program Outcome 6—"Demonstrate ethical and professional responsibility in engineering practice"—was conducted based on the evaluation of submitted OJT portfolios, which served as the primary data source. These portfolios contained required documentation such as daily journals, accomplishment reports, certificates of completion, and supervisor evaluation forms.

Results showed that the target performance level was successfully attained for both Performance Indicator 1 (Demonstrate ethical and professional behavior during practice) and Performance Indicator 2 (Demonstrate professionalism by complying with relevant codes and standards). A significant majority of the students received ratings between 4 and 5 across all indicators, indicating strong evidence of outcome achievement. This is further reflected in the overall rating of 98%, confirming that Program Outcome 6 was met during the period of data collection.

It is worth noting that the structured format of the OJT portfolio and the consistent coordination with industry supervisors contributed positively to the reliability of the data. However, variations in the quality of supervision and the nature of tasks assigned by different host companies may have introduced some degree of variability in the students' learning experiences and evaluations. Nonetheless, the assessment results indicate that, overall, students effectively demonstrated the expected level of ethical and professional conduct in their respective training environments ([Attachment C9-25 – SO 6 Assessment Result](#)).

Results of Direct Assessment for SO 7

Communicate effectively on complex civil engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

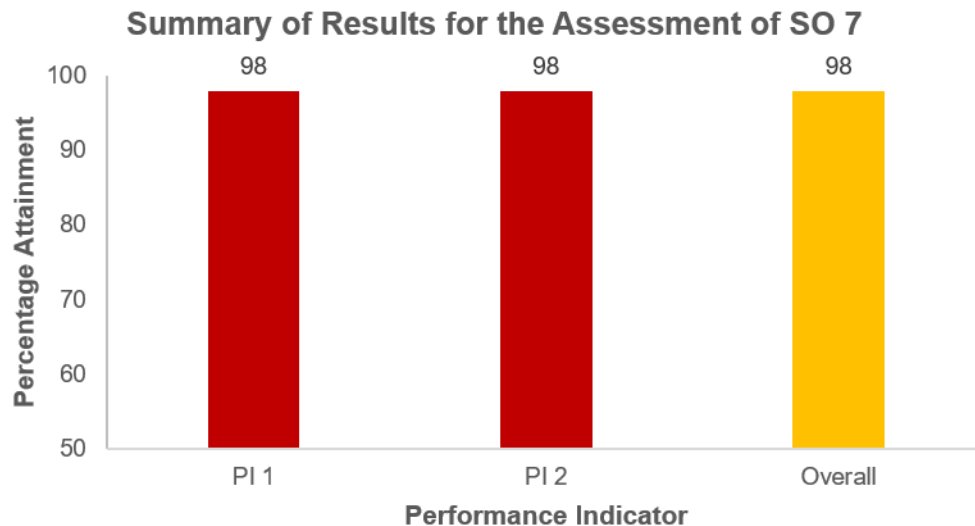


Figure 9.14 SO 7 Summary Result

For the 1st Semester of SY 2021–2022, assessment data was gathered from a total of 214 students enrolled across all sections of the Civil Engineering Project 1 subject. The evaluation of Student Outcome 7 (SO 7), which focuses on effective communication, was conducted using course-embedded assessment tools. Specifically, the Civil Engineering Project 1 outputs and the rubric-based evaluation of oral defenses in Civil Engineering Project 2 were utilized to measure performance indicators. Results showed that the target percentage was attained in both PI-1 (demonstrate effective written communication through grammatically correct and logically structured reports with appropriate use of graphics and technical style) and PI-2 (demonstrate effective oral communication through the clear articulation and defense of ideas supported by sound reasoning). A majority of students received performance ratings between 3 and 5 across both indicators. The high overall attainment rate of 98% clearly reflects that Student Outcome 7 was successfully achieved during the period. It is worth noting that the collaborative nature of project development and the formal oral defense process may have positively influenced the results by providing students with authentic opportunities to apply and demonstrate their communication skills in both written and oral formats ([Attachment C9–26 – SO 7 Assessment Result](#)).

Results of Direct Assessment for SO 8

Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex civil engineering problems in a global, economic, societal and environmental context.

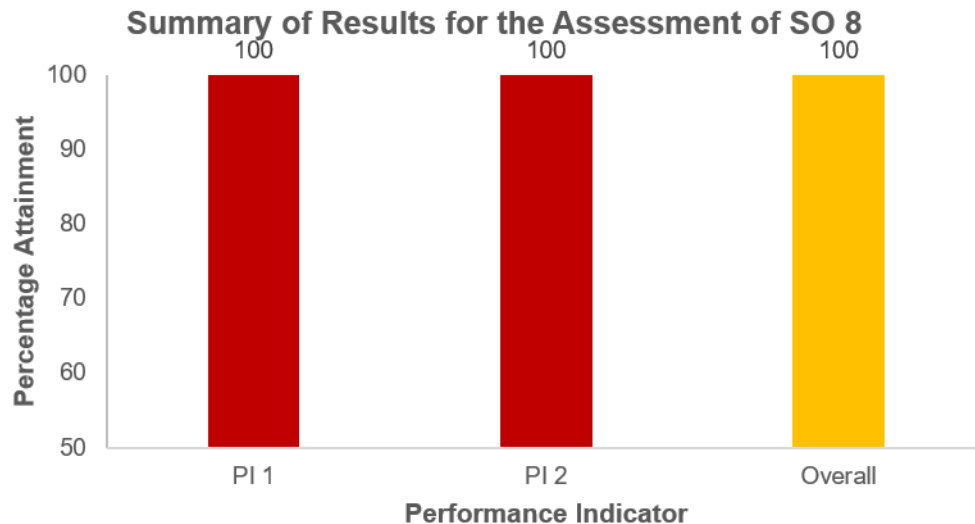


Figure 9.15 SO 8 Summary Result

For the 2nd Semester of SY 2021–2022, a total of 143 students across all sections were assessed in the Civil Engineering Project 2 subject for Student Outcome 8. The assessment focused on two performance indicators: PI-1 (Explain the impacts of civil engineering solutions in a global, economic, environmental, and societal context) and PI-2 (Establish the significance of the solutions in terms of beneficiaries). Data were gathered through the CE Project 2 Rubrics for Research Paper, which was used consistently across all sections. This rubric evaluates the students' ability to contextualize their proposed civil engineering solutions within broader societal and environmental frameworks and to articulate their relevance to various stakeholders.

The data reflect that the majority of students obtained ratings of 4 to 5 in both performance indicators, demonstrating a high level of proficiency. The overall attainment rate of 100% indicates that all students met or exceeded the minimum expected level of performance for this outcome. However, it is worth noting that certain factors may have influenced the results, including the modality of instruction, which was still largely shaped by adjustments to blended or flexible learning during the pandemic recovery period. Additionally, the collaborative nature of CE Project 2—where students work in teams to develop and present comprehensive research outputs—may have positively influenced individual performance ratings due to peer support and shared accountability. Faculty calibration sessions also helped ensure uniform application of the rubrics, thereby contributing to the reliability of the assessment results.

Overall, the data provide strong evidence that Student Outcome 8 was successfully attained during the period under review ([Attachment C9-27 – SO 8 Assessment Result](#)).

Results of Direct Assessment for SO 9

Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

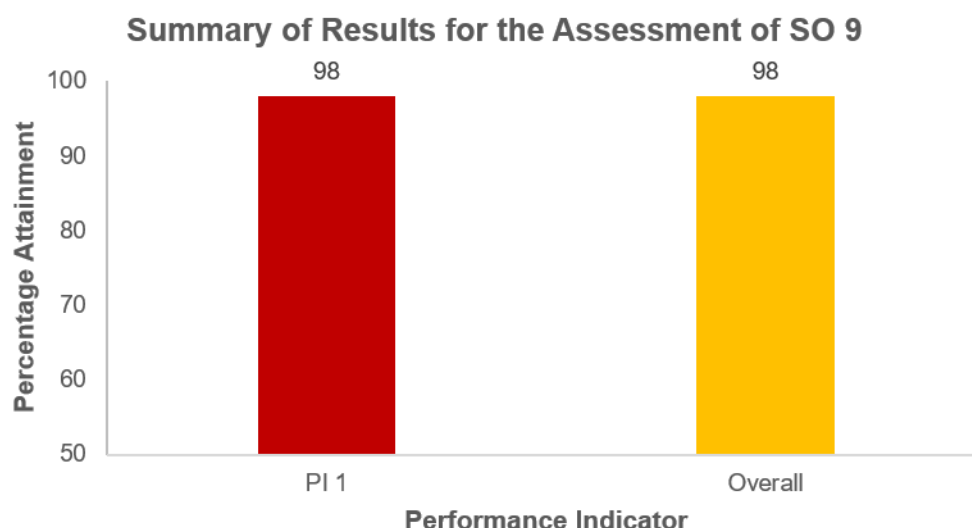


Figure 9.16 SO 9 Summary Result

For the 2nd Semester of SY 2021–2022, assessment data for Student Outcome 9 (Recognize the need for, and engage in lifelong learning) was gathered from 233 students enrolled across all sections in the Civil Engineering Fieldtrip course. The primary tools used were the Rubrics for the Assessment of Student Reports on Site Inspection and the Rubrics for the Assessment of Student Reports on Seminars, which were designed to evaluate students' ability to acquire new knowledge through various modes of learning and experiences during the fieldtrip activities.

These rubrics measured performance indicators such as students' initiative in exploring technical content, quality of insights gained from industry exposure, and reflections on professional growth. The evaluation was conducted by faculty members who facilitated the fieldtrip and assessed students' submitted reports and presentations.

Despite logistical challenges such as varying schedules of site visits, differences in the scope of industry partners' presentations, and limited time for post-activity reflections, the data collected remained consistent and reliable due to the uniform rubric criteria and calibration among assessors.

The results revealed that the target percentage was attained in PI-1 (The student can demonstrate self-directed learning through acquired new knowledge from various modes and strategies). A significant majority of students obtained ratings of 4 to 5 across the indicators, resulting in an impressive overall attainment of 98%. These findings affirm that Program

Outcome 9 was effectively achieved through the experiential and reflective components embedded in the Civil Engineering Fieldtrip course ([Attachment C9-28 – SO 9 Assessment Result](#)).

Results of Direct Assessment for SO 10

Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex civil engineering problems.

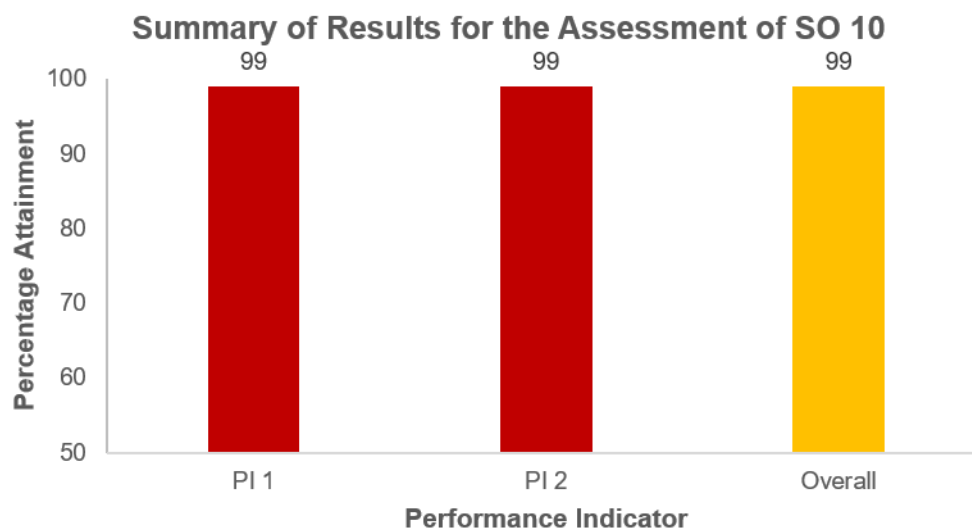


Figure 9.17 SO 10 Summary Result

For the 2nd Semester of SY 2020–2021, a total of 205 students across all sections enrolled in the course Civil Engineering Laws, Ethics, and Contracts were assessed for the attainment of Student Outcome 10 (SO 10). This outcome focuses on the ability to recognize the impact of contemporary issues in civil engineering practice. The assessment covered two performance indicators: PI-1 (Identify contemporary issues in a chosen civil engineering field of specialization) and PI-2 (Determine the effect of contemporary issues on the civil engineering discipline).

Data collection was carried out using a locally developed assessment tool, specifically designed to align with course outcomes and program-level competencies. The instrument included situational and scenario-based items that required students to demonstrate ethical reasoning, awareness of professional responsibilities, and understanding of the broader implications of engineering decisions.

To ensure objectivity and consistency, instructors administered the exam under standard conditions during regular class schedules. The majority of the students obtained

ratings between 4 and 5, indicating a high level of proficiency across the assessed indicators. The overall rating of 99% clearly demonstrates that the target percentage for SO 10 was successfully achieved during this cycle.

It is worth noting that the high attainment could also be attributed to increased awareness among students of current issues affecting the profession, such as sustainability, climate resilience, and regulatory reforms—topics that were actively integrated into class discussions and case study analyses. However, external factors such as varying instructional delivery methods (i.e., hybrid or flexible learning setups) and students' differing levels of exposure to real-world civil engineering contexts may have influenced the results to some extent.

Nonetheless, the assessment results provide strong evidence that the program has effectively instilled in students the ability to recognize and evaluate contemporary issues relevant to the civil engineering profession ([Attachment C9-29 – SO 10 Assessment Result](#)).

Results of Direct Assessment for SO 11

Create, select and apply appropriate techniques, skills, resources, and modern engineering and IT tools, including prediction and modelling, to complex civil engineering problems with an understanding of the limitations.

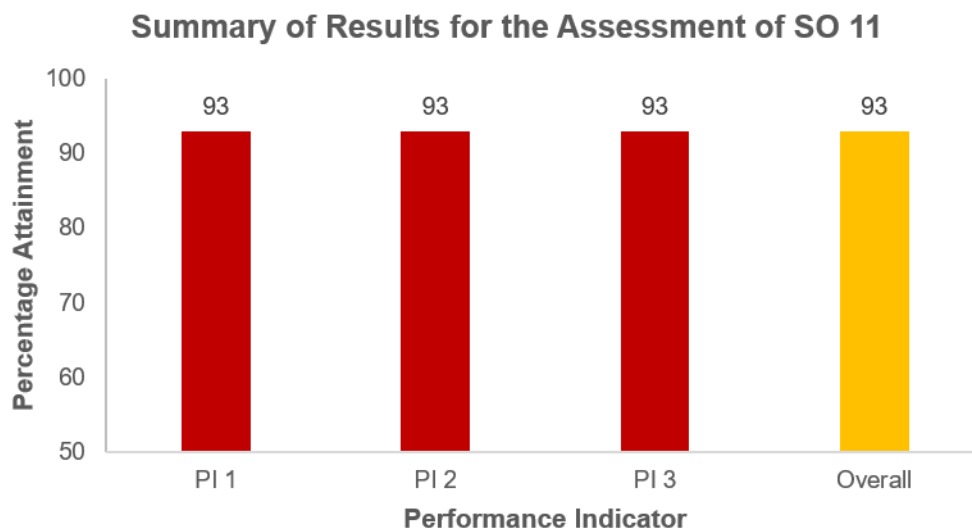


Figure 9.18 SO 11 Summary Result

For the 1st Semester of SY 2021–2022, a total of 208 students across all sections were assessed in the Reinforced Concrete Design course to evaluate the attainment of Student Outcome (SO) 11: "Recognize the need for, and engage in independent and life-long learning in the context of technological change." The data was primarily gathered through direct

assessment using the students' individual design plates, which were evaluated using a standardized Rubric Assessment for Design. These plates required students to apply theoretical concepts in the creation, selection, and application of appropriate design methods and tools aligned with modern civil engineering practices.

The assessment focused on three performance indicators:

- **PI-1:** Creation of the most appropriate techniques or modern tools for a specific civil engineering task,
- **PI-2:** Selection of skills relevant to civil engineering practice using modern engineering tools and techniques, and
- **PI-3:** Application of the techniques, resources, skills, and modern engineering tools to address complex civil engineering problems.

The majority of the students demonstrated proficiency by obtaining scores of 3 to 4 across all performance indicators, indicating satisfactory to very satisfactory performance levels. The consolidated data yielded an overall attainment rating of 93%, signifying that the program outcome was successfully met under this subject.

It is worth noting that several factors may have influenced the performance outcomes. The implementation of hybrid or blended learning modes during the semester may have posed limitations in hands-on experience and immediate feedback, especially when using specialized software and engineering tools. Additionally, varying levels of student access to design software, computing devices, and internet connectivity could have affected the quality and completeness of the submitted outputs. Despite these challenges, the consistent use of a rubric-based evaluation system ensured fairness and standardization in assessing the students' capabilities.

The assessment results highlight the effectiveness of the Reinforced Concrete Design course in cultivating the students' ability to independently select and apply modern tools and techniques, which are essential in solving real-world civil engineering problems in a constantly evolving technological landscape ([Attachment C9-30 – SO 11 Assessment Result](#)).

Results of Direct Assessment for SO 12

Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

CE Project 1

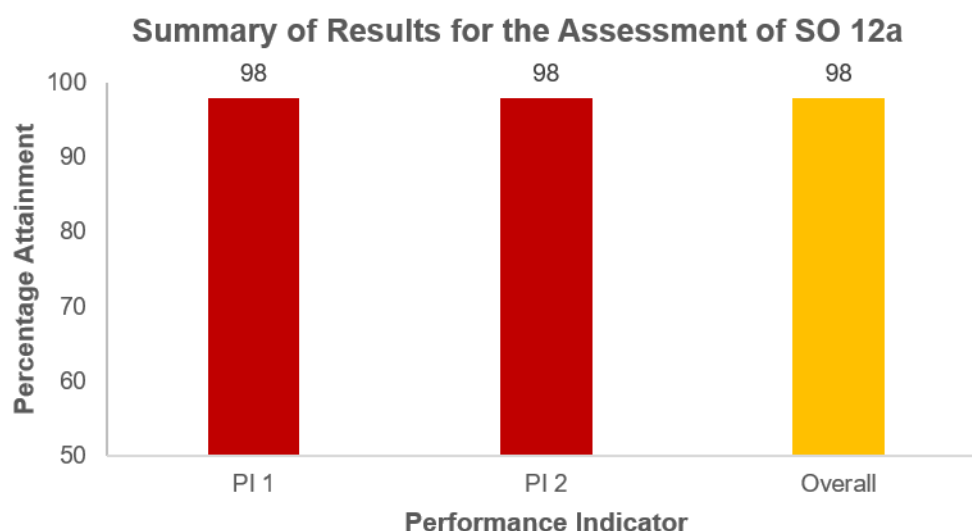


Figure 9.19 SO 12a Summary Result

For the 1st Semester of SY 2021–2022, data was gathered from 144 students enrolled across all sections of the Civil Engineering Project 1 subject to assess the attainment of Student Outcome 12 (SO12), which focuses on functioning effectively in a team to manage projects and solve real-world problems. The assessment utilized the rubrics for evaluating the research paper component of Civil Engineering Project 2, which allowed for a comprehensive evaluation of students' capabilities in applying appropriate tools and theories, as well as identifying and implementing solutions as part of a multidisciplinary team.

The data collection was based on direct faculty assessments using predefined performance indicators:

- PI-1: Use appropriate tools and theories to manage multidisciplinary projects in a team;
- PI-2: Identify and devise solutions to real-world problems as a member/leader of a cross-functional team.

The majority of the respondents received a rating of 4 out of 5 in both indicators, signifying a high level of performance. The high overall attainment rating of 98% confirms that the program outcome was successfully achieved for the given term.

It is important to note that during this period, academic activities were still being conducted in a flexible learning setup due to the ongoing effects of the COVID-19 pandemic. This may have impacted team dynamics and access to collaborative tools or project resources. Nevertheless, the students were able to adapt and demonstrate effective teamwork, problem-solving, and project management competencies despite these

challenges. The use of digital platforms for communication and collaboration likely played a crucial role in maintaining the integrity of group-based project work.

The assessment results suggest that the instructional strategies, guidance from faculty, and structured use of the CE Project 2 rubrics contributed to the successful attainment of the targeted student outcome. This reflects the program's ability to effectively prepare students for multidisciplinary collaboration in professional settings, even under non-traditional learning conditions ([Attachment C9-31 – SO 12a Assessment Result](#)).

On-The-Job Training (OJT)

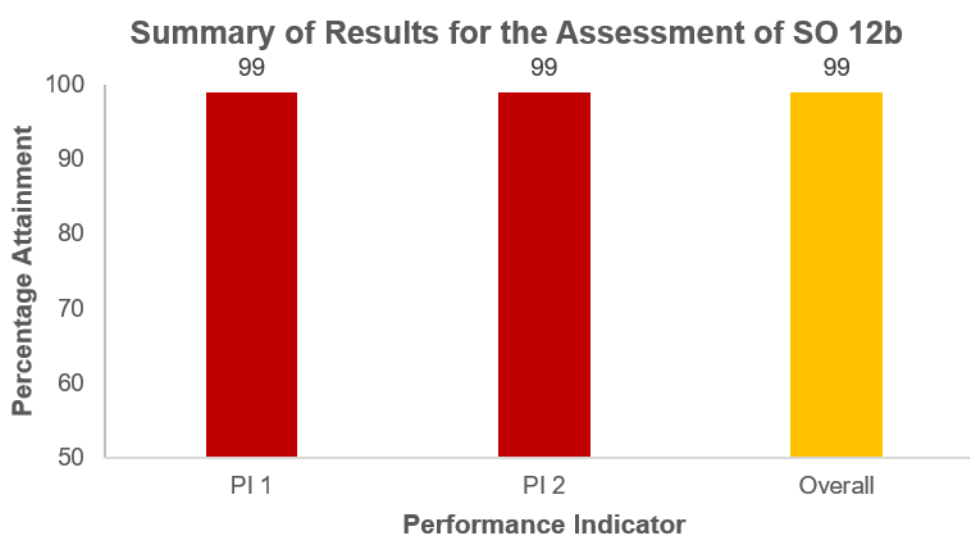


Figure 9.20 SO 12b Summary Result

For the Summer Term of SY 2020–2021, data were gathered from 215 students enrolled across all sections in the On-the-Job Training 1 (OJT 1) subject. The primary instrument used for assessment was the OJT Portfolio Performance Evaluation Rubrics, which measured student performance based on key indicators aligned with Student Outcome 12. Specifically, these were: PI-1 – Use appropriate tools and theories to manage multidisciplinary projects in a team, and PI-2 – Identify and devise solutions to real-world problems as a member/leader of a cross-functional team.

The data collection process involved the evaluation of each student's OJT portfolio, which included documentation of actual work experience, project contributions, and reflections during their internship. These were validated and rated by assigned faculty supervisors based on the criteria outlined in the rubrics.

A notable scenario that may have influenced the results was the gradual transition from remote to hybrid internship setups due to the easing of pandemic-related restrictions. While some students completed their training through in-person arrangements, others remained under remote or online industry-based engagements. Despite this variation in modality, students consistently demonstrated strong performance across all indicators

The results showed that a majority of the students obtained the highest rating (5) in both performance indicators, reflecting their ability to apply theoretical knowledge in professional practice and collaborate effectively within multidisciplinary teams. This is clearly supported by the high overall attainment rating of 99%, indicating that Student Outcome 12 was successfully met through the implementation of OJT 1 during the specified term ([Attachment C9-32 – SO 12b Assessment Result](#)).

Summary of SO Assessment Result

Table 9.7 presents the **Summary of Student Outcomes (SO) Assessment** for the Civil Engineering program of the Polytechnic University of the Philippines (PUP). Based on the results, all identified Student Outcomes were marked as **Attained**, indicating that the program was able to meet its expected learning outcomes across the assessed courses and related academic activities.

The results show consistently high attainment ratings, ranging from 79% to 100%, which reflects the department's strong implementation of outcome-based education. Several Student Outcomes registered very high ratings. SO 5 and SO 8, both assessed through CE Project 2, obtained the highest rating of 100%, showing excellent student performance in these outcome areas. Similarly, SO 1 in RCD, SO 2 in CE Research, and SO 10 in CE Laws, Ethics and Contracts each posted 99% attainment, indicating that students demonstrated strong competence in the knowledge, research, and professional responsibility components of the program.

Other Student Outcomes also reflected very strong performance. SO 4 and SO 6, both assessed through OJT 2, obtained 98% attainment, suggesting that students were able to apply their learning effectively in practical and field-based environments. SO 7, assessed in CE Project 1, also reached 98%, while SO 9, assessed through CE Fieldtrip, similarly achieved 98%, indicating that experiential and project-based learning activities significantly contributed to the development of the expected competencies. SO 11, assessed in RCD, posted 93% attainment, which still reflects a strong level of achievement. For SO 12, assessment through CE Project 1 and OJT yielded 98% and 99%, respectively, further confirming students' high level of competence in the intended outcome.

Overall, the summary of results demonstrates that the Civil Engineering program of PUP has achieved a high level of Student Outcomes attainment. The consistently strong ratings

imply that the curriculum, instruction, laboratory and field activities, research exposure, and practicum experiences are effectively contributing to student learning and competency development. These results also affirm that the department is successfully preparing its students in accordance with the program's educational goals and the demands of the civil engineering profession.

Table 9.7 Summary of SO Assessment

SO Number	Subject Assessed	Attainment Rating	Remarks
1	RCD	99%	Attained
2	CE Research	99%	Attained
3	CE Project 2	79%	Attained
4	OJT 2	98%	Attained
5	CE Project 2	100%	Attained
6	OJT 2	98%	Attained
7	CE Project 1	98%	Attained
8	CE Project 2	100%	Attained
9	CE Fieldtrip	98%	Attained
10	CE Laws, Ethics and Contracts	99%	Attained
11	RCD	93%	Attained
12	CE Project 1	98%	Attained
	OJT	99%	Attained

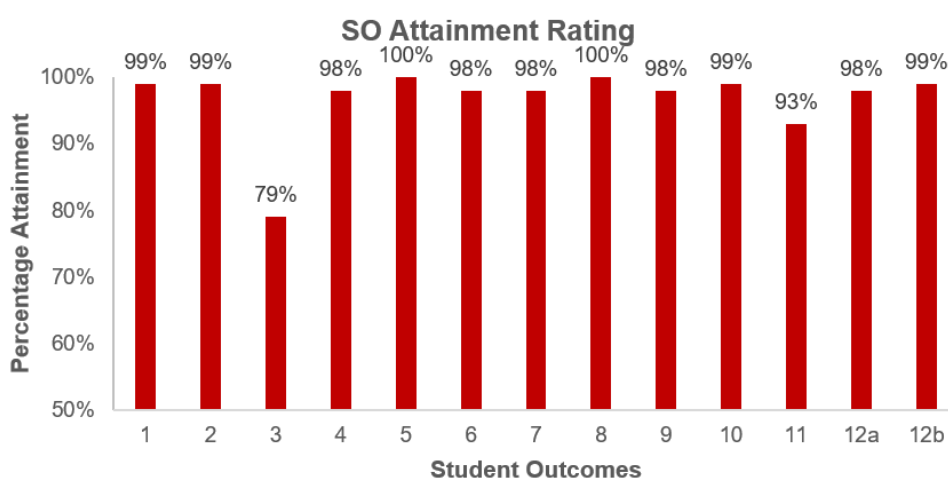


Figure 9.21 Summary of SO Assessment

C. Continuous Quality Improvement

Describe how the results of evaluation processes for the program educational objectives and the student outcomes and any other available information have been used as input in the continuous improvement of the program. Provide a list of continuous improvement actions taken and include results of the actions taken. Likewise, indicate any significant future program improvement plans based upon recent evaluations. Provide a brief rationale for each of these planned changes.

The Civil Engineering Department at PUP systematically uses the results of evaluation processes for both Program Educational Objectives (PEOs) and Student Outcomes (SOs), along with other relevant information, as critical input for the continuous quality improvement of the program. Data are collected through employer and alumni surveys, student course evaluations, licensure examination performance, internship feedback, and faculty assessments. These results are analyzed by the Continuous Quality Improvement (CQI) Team, then deliberated in Department and Advisory Board meetings to identify responsive actions. Among the significant improvements undertaken were the enhancement of course syllabi to strengthen the alignment of Student Outcomes with course learning outcomes, which ensured consistency in outcome mapping and clarity of assessment rubrics. Faculty development initiatives, such as training and workshops, were also implemented to address gaps in teaching effectiveness, resulting in improved student feedback on instructional delivery. Industry linkages were expanded to provide more robust internship opportunities and industry-based projects, which led to higher employability and positive employer feedback regarding graduate preparedness. In addition, research and extension activities were realigned to focus on sustainability and disaster resilience in response to alumni and industry recommendations. The results of these actions have been evident in higher attainment levels of SOs, improved internship and employer satisfaction ratings, and enhanced licensure examination performance in recent years.

Building on these outcomes, the department has identified several significant future improvement plans based on recent evaluations. These include the development of a centralized digital database for the systematic storage and tracking of assessment results to strengthen evidence-based decision-making and accreditation reporting, and the expansion of employer and alumni tracer surveys to cover a broader base of stakeholders, thereby increasing the validity of PEO attainment evaluations. Likewise, plans are underway to enhance the integration of sustainability-focused, multidisciplinary projects into the capstone design experience in order to further strengthen holistic and future-ready engineering competencies. Finally, the Department seeks to expand faculty exchange and industry immersion programs, ensuring that instruction and research remain aligned with current

industry practices. These planned initiatives are supported by the rationale of keeping the program responsive to stakeholder needs and ensuring continued compliance with both national and international standards of engineering education.

Based on the results of the Program Educational Objectives (PEO) and Student Outcomes (SO) assessment, the Civil Engineering Department of the Polytechnic University of the Philippines (PUP) initiated several Continuous Quality Improvement (CQI) actions to address gaps in instruction, laboratory delivery, curriculum implementation, research development, industry exposure, and learning resources. These initiatives were designed to strengthen student competence, improve academic delivery, and ensure that graduates remain responsive to professional and societal needs as shown in Table 9.8.

1. Creation of Instructional Modules

One of the major CQI actions undertaken by the department was the creation of instructional modules for key Civil Engineering courses. This initiative was introduced to provide students with more organized, structured, and accessible learning materials aligned with course outcomes and program outcomes. The availability of instructional modules improved lesson consistency across course sections and helped students better understand fundamental and advanced engineering concepts. It also supported independent learning and improved student preparedness for classroom discussions, quizzes, and major examinations.

2. Creation of Laboratory Manuals

The department also developed laboratory manuals to standardize laboratory exercises, procedures, expected outputs, and safety practices. This action was in response to the need for more outcome-based laboratory instruction and improved hands-on learning experiences. The laboratory manuals enhanced the quality and consistency of laboratory activities by giving students clear procedures, objectives, and report formats. This strengthened students' practical skills, experimental competency, and technical reporting ability. It also improved the assessment of laboratory-based student outcomes, especially those related to experimentation, data interpretation, and the use of engineering tools and techniques.

3. Revision of Syllabus

Another significant CQI initiative was the revision of course syllabi to ensure alignment with updated program outcomes, current engineering practices, and institutional goals. The revision included refinement of course content, learning outcomes, teaching strategies, and assessment methods. The revised syllabi contributed to better curriculum alignment with the

PEOs and student outcomes. Faculty members were guided more clearly in implementing outcome-based education, while students benefited from more focused and relevant course content. It also helped the department ensure that graduates are equipped with competencies expected in the civil engineering profession.

4. Course CQI Direct Survey

The department implemented a Course CQI Direct Survey to gather evidence from direct assessment methods such as examinations, projects, laboratory outputs, design work, and performance tasks. This initiative aimed to identify specific areas in which students were meeting or falling short of the expected learning outcomes. The direct survey provided measurable evidence of student performance and helped faculty identify weak competencies that needed intervention. This led to more data-driven instructional decisions and a stronger culture of evidence-based quality assurance within the department.

5. Course CQI Indirect Survey

In addition to direct measures, the department conducted a Course CQI Indirect Survey to capture student perceptions regarding course effectiveness, learning experiences, and self-assessed attainment of outcomes. The indirect survey gave the department valuable feedback on how students viewed their learning experiences, classroom engagement, and difficulties encountered in various courses. This helped validate direct assessment results and revealed concerns that may not be immediately visible through grades alone. The initiative also encouraged greater student participation in quality improvement processes.

6. Creation of the Department's Research Agenda and Manual

To strengthen the research capability of both students and faculty, the department initiated the creation of a departmental research agenda and manual. This was intended to guide research directions toward relevant civil engineering issues and promote a more systematic approach to undergraduate and faculty research. The establishment of a research agenda improved the coherence and relevance of research topics undertaken in the department. It provided direction for thesis and research activities that are aligned with the UN Sustainable development goals (UN SDGs), national development needs, industry trends, and community concerns. The research manual also clarified procedures, standards, and expectations, resulting in more organized and quality research outputs. This initiative contributed to the development of analytical, investigative, and innovation-related student outcomes.

7. Enhancement of OJT Deployment Policies and Activities

Recognizing the importance of industry immersion, the department enhanced its On-the-Job Training (OJT) deployment policies and activities. This action aimed to improve student exposure to real engineering environments and ensure that practicum experiences are aligned with program objectives. The improved OJT policies resulted in more meaningful field exposure and better coordination with partner agencies and companies. Students gained a stronger practical understanding of professional engineering work, workplace ethics, communication, teamwork, and safety practices. The initiative also helped strengthen the connection between academic preparation and industry expectations, thereby contributing positively to the employability and professional readiness of graduates, which supports the attainment of the PEOs.

8. Additional Computer Software

The department also responded to assessment results by providing additional computer software relevant to civil engineering instruction, analysis, design, drafting, and simulation. The availability of additional software enhanced students' exposure to modern engineering tools and digital applications used in professional practice. This improved their technical competence in design, modeling, analysis, and presentation of engineering solutions. It also helped faculty integrate technology more effectively into instruction. As a result, students became better prepared for industry requirements and for outcomes involving modern tool usage, design competency, and professional practice readiness.

9. Additional Laboratory Equipment and Apparatus Procured and Requested

To improve practical training, the department procured and requested additional laboratory equipment and apparatus. This CQI action was undertaken to address limitations in laboratory resources and to support more effective experimental learning. The added laboratory equipment improved the conduct of experiments, demonstrations, and testing activities in various Civil Engineering subjects. Students were provided with better opportunities to perform actual measurements, testing procedures, and material evaluations, which strengthened experiential learning. This contributed to improved laboratory performance, enhanced technical confidence, and better attainment of outcomes related to experimentation, investigation, and engineering practice.

10. Encourage Alumni Members to Donate Equipment and Apparatus

Another important CQI initiative of the Civil Engineering Department was to encourage alumni members to donate laboratory equipment and apparatus to support the academic and

practical training needs of students. This initiative was driven by the department's recognition that alumni, as products of the program and now part of the professional community, can play a valuable role in strengthening the learning environment of future Civil Engineering students. Since some alumni of the department have already donated equipment and apparatus, this initiative also reflects the strong continuing relationship between the department and its graduates. The donations received from alumni contributed to the enhancement of laboratory instruction by increasing the availability of equipment used in demonstrations, experiments, and practical exercises. These additional resources helped address limitations in laboratory facilities and supported more effective hands-on learning experiences for students. The initiative also reduced, to some extent, the resource gaps encountered in conducting laboratory classes and technical activities. Beyond the material benefit, the action strengthened alumni engagement and reinforced a culture of shared responsibility for the continuous development of the department. It also showed students that graduates of the program remain committed to giving back and supporting the advancement of Civil Engineering education in PUP.

Table 9.8 Listing of CQI Initiatives and Actions Taken

Initiatives	Period Initiated	Remarks
1. Creation of Instructional Modules	SY 2024-2025	On-going updates
2. Creation of Laboratory Manuals	SY 2023-2024	Done
3. Revision of Syllabus	SY 2024-2025	Annual revision
4. Course CQI Direct Survey	SY 2024-2025	Annual survey
5. Course CQI Indirect Survey	SY 2024-2025	Annual survey
6. Creation of Department's Research Agenda and Manual	SY 2024-2025	Done
7. Enhancement of OJT Deployment Policies and Activities	SY 2024-2025	Done
8. Additional Computer Software	SY 2024-2025	Done
9. Additional Laboratory Equipment/Apparatus procured and Requested	SY 2024-2025	On-going updates
10. Encourage Alumni members to donate Equipment and Apparatus	SY 2024-2025	On-going

Future Program Improvement Plans

Based on the results of the Program Educational Objectives (PEO) and Student Outcomes (SO) assessments, the Civil Engineering Department has demonstrated satisfactory overall attainment. However, the assessment results also identified areas with comparatively lower performance. These findings will serve as the basis for the department's future improvement initiatives under its Continuous Quality Improvement (CQI) process.

1. Strengthening Industry and Alumni Engagement Programs

The department will strengthen its engagement with alumni and industry partners through more structured activities, such as annual alumni forums, mentoring programs, technical talks, industry consultations, and career development sessions. Employer feedback mechanisms will also be expanded and incorporated more consistently into curriculum review and program evaluation activities. These initiatives will provide the department with updated information on industry expectations, graduate performance, and emerging trends in civil engineering practice.

2. Institutionalization of Professional Organization Participation

Student membership and active participation in recognized professional engineering organizations will be encouraged through appropriate incentives and integration into co-curricular activities. Participation in these organizations will provide students with opportunities for professional networking, leadership development, technical learning, and community involvement. The department will also promote community-based engineering projects through selected courses, student activities, and extension programs.

3. Enhancement of Lifelong Learning and Continuing Education Pathways

The department will explore the development of short courses, micro-credentials, certification programs, and other continuing education activities for students, graduates, and practicing professionals. Linkages with graduate schools, professional institutions, government agencies, and training providers will also be strengthened to provide clearer pathways for advanced studies, professional certification, and continuing professional development.

4. Integration of Emerging Technologies in Curriculum

Relevant applications of artificial intelligence, data analytics, building information modeling, and advanced engineering software will be integrated into appropriate courses. The department will also work toward improving laboratory facilities, software availability, internet

connectivity, and computational resources. These improvements will allow students to gain practical experience using modern tools and technologies currently applied in the civil engineering profession.

5. Expansion of Experiential and Work-Integrated Learning

On-the-Job Training will be strengthened through more structured partnerships with construction companies, consulting firms, government agencies, and other related organizations. The department will improve the implementation of OJT by establishing clearer learning outcomes, closer monitoring procedures, and systematic evaluation of student performance. Whenever feasible, capstone and design projects will be undertaken in coordination with industry partners or actual clients to expose students to real engineering problems and professional expectations.

6. Enhancement of Communication and Professional Skills Training

Technical writing, oral presentation, teamwork, and interdisciplinary communication will be further reinforced across relevant courses. These skills will be integrated into course requirements, laboratory reports, design projects, and research presentations. The department will also conduct regular seminars on engineering ethics, contracts, project documentation, workplace responsibilities, and professional practice to better prepare students for employment and professional licensure.

7. Strengthening Sustainability and Social Impact Integration

Sustainable design principles, environmental impact assessment, disaster resilience, resource efficiency, and climate-responsive engineering will be given greater emphasis in major design and professional courses. The department will also promote community-based infrastructure projects that respond to identified local needs. These activities will help students understand the environmental, economic, and social responsibilities associated with civil engineering practice.

8. Development of a Centralized Digital CQI Database

The department plans to develop a centralized digital database for the management of PEO and SO assessment results, survey responses, action plans, implementation records, and supporting documents. The database will improve the organization, retrieval, and monitoring of assessment information. It will also support longitudinal analysis and allow the department to identify trends, monitor action plans, and make timely decisions based on available evidence.

9. Continuous Faculty Development Programs

Faculty development activities will be sustained through regular training and workshops on outcomes-based education, assessment methods, instructional strategies, research, technology integration, and current civil engineering practices. Faculty members will also be encouraged to participate in professional training, industry immersion, research activities, conferences, and innovation projects. These efforts will help ensure that teaching practices and course content remain relevant and responsive to the needs of students and the profession.

10. Expansion of Assessment Coverage and Stakeholder Participation

The department will improve the coverage of its assessment activities by increasing the response rates of alumni, employers, students, and other stakeholders. Participation from industry experts, professional organizations, government agencies, and community partners will also be expanded during surveys, consultations, curriculum reviews, and program evaluations. Broader stakeholder participation will provide more representative feedback and strengthen the evidence used in the department's CQI process.

D. Maintenance of Continuous Quality Improvement Program

Discuss how the Continuous Quality Improvement program is maintained including a description of how records are kept, and a discussion of resources needed for the maintenance of the program.

The Civil Engineering Department at PUP is committed to maintaining a Continuous Quality Improvement (CQI) program that ensures the delivery of high-quality education. This program is underpinned by a systematic approach to assessment, evaluation, and improvement.

The Civil Engineering Department maintains a systematic process for assessing program performance, documenting results, and implementing appropriate improvement actions. These practices support the department's Continuous Quality Improvement (CQI) process and provide evidence for evaluating the effectiveness of the curriculum, instructional delivery, student learning, and other program activities.

Assessment and Evaluation

1. Program-Level Outcomes Assessment

The department regularly assesses the attainment of its Program Educational Objectives (PEOs) and Student Outcomes (SOs) using both direct and indirect assessment methods. These include alumni surveys, employer evaluations, stakeholder feedback, and

summative assessments of student performance. The results are reviewed to determine whether the established targets have been achieved and to identify areas that require further improvement.

2. Course-Level Assessment

Faculty members use various assessment methods, including examinations, laboratory activities, design projects, research outputs, presentations, and other course requirements, to evaluate student performance in relation to the prescribed course learning outcomes. The assessment results are documented and analyzed to determine the extent to which students have achieved the expected competencies. Findings from course-level assessments are also used to improve teaching strategies, course content, learning activities, and assessment tools.

3. Preparation of Self-Study Reports

Periodic self-study reports are prepared to provide a comprehensive evaluation of the Civil Engineering program. These reports examine the department's strengths, weaknesses, opportunities, and threats and present relevant information on curriculum implementation, faculty qualifications, facilities, student performance, stakeholder feedback, and program outcomes. The results of the self-study serve as a basis for identifying priorities and developing appropriate improvement measures.

4. Research-Based Assessment

The department, through its faculty members and students, conducts research-based assessments to evaluate the effectiveness and efficiency of its academic and administrative activities. These studies may cover areas such as On-the-Job Training implementation, curriculum delivery, instructional practices, student performance, and other curricular concerns. The findings are used to support evidence-based decisions and to recommend improvements in program policies, procedures, and activities.

Record Keeping

1. Maintenance of Assessment Data

Assessment records, including student performance data, survey results, evaluation reports, course assessment documents, and PEO and SO attainment results, are properly maintained by the department. Electronic copies are stored in the department's designated Google Drive, which serves as its current digital repository. Files are organized to facilitate retrieval, review, monitoring, and presentation during internal and external evaluations.

2. Documentation of Meeting Minutes

Minutes of department meetings, faculty meetings, committee meetings, consultations, and other relevant activities are prepared and archived. These records contain the matters discussed, decisions made, assigned responsibilities, recommended actions, and agreed timelines. The meeting minutes provide evidence of stakeholder participation and serve as a historical record of the department's decisions and actions.

3. Documentation and Monitoring of Improvement Plans

Improvement plans resulting from assessment findings, stakeholder feedback, and program reviews are properly documented. Each plan identifies the concern to be addressed, the proposed action, the responsible persons or committees, the implementation period, and the expected output. Progress reports and supporting documents are maintained to monitor the status and effectiveness of CQI initiatives.

Continuous Improvement

1. Data-Driven Decision-Making

The department uses the results of assessments, surveys, evaluations, and consultations to identify patterns, strengths, weaknesses, and areas requiring attention. These findings provide the basis for decisions concerning curriculum enhancement, instructional delivery, faculty development, student support, facilities, and other program needs. Improvement initiatives are prioritized according to the significance of the findings and the availability of institutional resources.

2. Regular Review and Adjustment

The department's CQI processes, policies, procedures, and assessment practices are reviewed regularly to ensure their continued relevance and effectiveness. Necessary adjustments are made based on assessment results, stakeholder recommendations, changes in professional practice, institutional policies, and accreditation requirements. The implementation and outcomes of improvement actions are also reviewed to determine whether the identified concerns have been adequately addressed.

3. Collaboration and Partnerships

The department maintains collaboration with industry partners, alumni, professional organizations, government agencies, and other academic institutions. These partnerships provide opportunities for consultation, resource sharing, professional development, research

collaboration, student training, and the exchange of good practices. Inputs from external stakeholders are considered in curriculum reviews, program evaluations, and the development of improvement initiatives.

Through these assessments, documentation, and continuous improvement practices, the Civil Engineering Department remains committed to providing quality engineering education that responds to the needs of its students, stakeholders, and the civil engineering profession. The department will continue to strengthen its CQI system to ensure that program decisions and improvement actions are supported by appropriate evidence and regularly monitored for effectiveness.

Individual Course Continuous Quality improvement

Starting School Year 2025–2026, the Civil Engineering Department of Polytechnic University of the Philippines has strengthened its commitment to Outcomes-Based Education (OBE) by implementing a systematic Individual Course Continuous Quality Improvement (CQI) process. This initiative aims to regularly assess and enhance the effectiveness of teaching and learning within the current curriculum, ensuring alignment with national and international accreditation standards ([Attachment C9-33 Individual Course CQI Form](#)).

Under this framework, all courses offered by the department are subjected to structured evaluation every semester. Faculty members assigned to each subject are responsible for assessing the attainment of Course Intended Learning Outcomes (CILOs) using two complementary approaches: direct and indirect assessment methods ([Attachment C9-34 Sample Google Form for Indirect Survey](#)).

The direct assessment method is based on measurable student performance, primarily derived from the final grades obtained at the end of the semester. These grades serve as quantitative indicators of how well students have achieved the expected learning outcomes in each course. On the other hand, the indirect assessment method captures students' perceptions of their learning through structured surveys administered via Google Forms. These surveys focus on students' self-assessment of their attainment of the CILOs, providing valuable qualitative insights into the teaching-learning process ([Attachment C9-35 Individual Course CQI Results](#)).

All collected data and analyses are submitted to the **CQI Committee Head**, who oversees the consolidation of results across courses. The CQI Committee then evaluates the findings, identifies trends, and formulates evidence-based recommendations aimed at improving instructional delivery, assessment strategies, and curriculum design.

The final consolidated CQI report is subsequently submitted to the Department Chairperson for further review. These results are also presented during faculty meetings to

ensure transparency, collaborative reflection, and shared accountability among faculty members.

Based on the recommendations, the Chairperson initiates appropriate actions, which may include instructional adjustments, faculty development programs, curriculum revisions, or coordination with relevant university offices for institutional support. These actions are geared toward addressing identified gaps and enhancing student learning outcomes. Ultimately, this CQI process is designed to systematically improve the percentage attainment of course outcomes, fostering a culture of continuous improvement and ensuring that the Civil Engineering program remains responsive, relevant, and compliant with accreditation standards ([Attachment C9-36 Summary of Individual Course CQI Assessment](#)).

E. Additional Information

Copies of any of the assessment instruments or materials must be available for review at the time of the visit. Other information such as minutes from meetings where the assessment results were evaluated and where recommendations for action were made could also be included.

In compliance with the requirement on the availability of assessment instruments and supporting materials, the program ensures that all relevant documentation is systematically compiled and maintained. Copies of the assessment instruments, rubrics, and other evaluation materials used in measuring student outcomes and program educational objectives are readily accessible. In addition, minutes of meetings where assessment results were reviewed, analyzed, and deliberated upon, together with corresponding recommendations for action, are properly recorded and archived. These materials serve as evidence of the program's commitment to continuous quality improvement and are available for review during the accreditation visit. All required additional documents will likewise be provided upon request by the evaluators.

The College has an appointed Quality Assurance (QA) Coordinator who oversees and helps manage the quality assurance functions of the college. The QA Coordinator plays a vital role in coordinating with the different academic programs regarding the implementation of Continuous Quality Improvement (CQI) initiatives. As part of this responsibility, the coordinator gathers, consolidates, and summarizes the findings of assessments, evaluations, and program reports submitted by the various departments. These consolidated evaluation results are then reported to the Dean of the College to serve as the basis for appropriate action and decision-making. Through this process, the proposed interventions are properly reviewed and supported to help improve the attainment of Student Outcomes (SO) and Program Educational

Objectives (PEO), thereby strengthening the overall quality and effectiveness of the academic programs.

In addition to the PEO and SO surveys, the Civil Engineering Department undertakes other assessment-related initiatives to further strengthen the evaluation and improvement of its academic program. The faculty members and students conduct research-based assessments focusing on the efficiency and effectiveness of various curricular activities implemented in the department. The findings from these studies are submitted to the Department Chair for validation and are subsequently discussed during faculty meetings and GIAB meetings for careful review, scrutiny, and possible implementation when necessary. Another important source of assessment information comes from faculty coordination meetings and committee meetings, where issues affecting curriculum implementation, instruction, and academic delivery are discussed in detail. These meetings provide valuable insights into operational and instructional concerns that may require intervention or adjustment. In addition, board examination results serve as a significant source of assessment data, as they provide information on the passing rates of graduates and reflect the department's performance in preparing students for professional licensure. Taken together, these initiatives provide multiple sources of evidence that support informed decision-making and continuous quality improvement in the Civil Engineering program.

ATTACHMENTS

[Attachment C9-01 – PEO Survey Form](#)

[Attachment C9-02 – Screenshot of the Google Link Form](#)

[Attachment C9-03 – The CQI Committee](#)

[Attachment C9-04 – The PEO and Its Indicators](#)

[Attachment C9-05 – PEO Result Report](#)

[Attachment C9-06 – Minutes of the Meeting discussing PEO Result\)](#)

[Attachment C9-07 – List of the 2022 Graduates](#)

[Attachment C9-08 – Alumni Survey Result](#)

[Attachment C9-09 – Employer Survey Result](#)

[Attachment C9-10 – Graduate Tracer Study 2022](#)

[Attachment C9-11 – Copies of the Research Papers on Assessment](#)

[Attachment C9-12 – Licensure Examination Data](#)

[Attachment C9-13 – MOM of Department Meetings](#)

[Attachment C9-14 – Department Minutes of the Meeting](#)

[Attachment C9-15 – MOM of Department's Meeting](#)

[Attachment C9-16 – RA 10533](#)

[Attachment C9-17 – Rubrics Forms](#)

[Attachment C9-18 – Sample of Indirect Assessment Survey Form](#)

[Attachment C9-19 – Sample Course Syllabi](#)

[Attachment C9-20 – SO 1 Assessment Result](#)

[Attachment C9-21 – SO 2 Assessment Result](#)

[Attachment C9-22 – SO 3 Assessment Result](#)

[Attachment C9-23 – SO 4 Assessment Result](#)

[Attachment C9-24 – SO 5 Assessment Result](#)

[Attachment C9-25 – SO 6 Assessment Result](#)

[Attachment C9-26 – SO 7 Assessment Result](#)

[Attachment C9-27 – SO 8 Assessment Result](#)

[Attachment C9-28 – SO 9 Assessment Result](#)

[Attachment C9-29 – SO 10 Assessment Result](#)

[Attachment C9-30 – SO 11 Assessment Result](#)

[Attachment C9-31 – SO 12a Assessment Result](#)

[Attachment C9-32 – SO 12b Assessment Result](#)

[Attachment C9-33 – Individual Course CQI Form](#)

[Attachment C9-34 – Sample Google Form for Indirect Survey](#)

[Attachment C9-35 – Individual Course CQI Results](#)

[Attachment C9-36 – Summary of Individual Course CQI Assessment](#)